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Products Liability in a World Designed for Men

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Automobile crashes are a routine feature of modern life, yet their consequences are not evenly distributed. Women driving most vehicles face substantially higher risks of serious injury and death than men in comparable crashes—a disparity linked to safety systems designed and tested around male bodies. Across a wide range of consumer products manufacturers continue to rely on male anthropometric and physiological data as the implicit design baseline, resulting in systematic disparities in safety outcomes for women.

Despite the prevalence and foreseeability of these harms, products liability law has never squarely confronted whether a product designed around male bodies, marketed as gender neutral, and foreseeably more dangerous for women constitutes a design defect. Prevailing design defect doctrine evaluates product safety in the aggregate, allowing strong performance for men to obscure systematic harms to women.

This Article identifies male-default design as an analytically distinct and underexamined source of design defect. It argues that existing products liability doctrine does not require courts to ignore subgroup harms where those harms affect a substantial and foreseeable class of ordinary users.

Building on this foundation, the Article develops a framework for incorporating subgroup considerations into design defect analysis. It argues, first, that courts should more consistently account for subgroup outcomes to prevent aggregate safety metrics from masking systematic harms to women. It then advances a more robust approach that treats persistent sex-based disparities in safety as independently relevant to defectiveness, particularly where products are marketed to both sexes but designed around male bodies. By reframing design defect doctrine to attend not only to aggregate safety but also to the allocation

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of risk among users, the Article seeks to realign products liability law with its core commitments to foreseeability, fairness, and deterrence.

INTRODUCTION

Car crashes are an unavoidable reality of modern life. The average American driver will be involved in three to four automobile accidents over the course of a lifetime.[2] Yet women face substantially worse outcomes in those crashes. A woman driving most cars is 47% more likely to suffer serious injury, 71% more likely to be moderately injured, and 17% more likely to die than a male driver in the same collision. [3]

A similar pattern appears in medicine. Americans spend nearly half their lives taking at least one prescription drug, and by age fifty most are taking five or more medications at once.[4] Yet those drugs are riskier for women. Women experience adverse drug reactions more often than men, including reactions serious enough to require hospitalization.[5]

[2] Karen Aho, *Here's How Many Car Accidents You'll Have*, Fox Business, *available at* https://www.foxbusiness.com/features/heres-how-many-car-accidents-youll-have (2016) (noting that car insurance companies find that the average American will file a claim for a car accident once every 17.9 years)

[3] Dipan Bose, Maria Segui-Gomez & Jeff R. Crandall, *Vulnerability of Female Drivers Involved in Motor Vehicle Crashes: An Analysis of US Population at Risk*, 101 AM J. PUBLIC HEALTH 2368, 2371 (2011); Patrick Miner et al., *Car Harm: A Global Review of Automobility's Harm to People and the Environment*, 115 J. TRANSPORT GEOGRAPHY 8 (2011). C.J. Kahane, NHTSA Injury *Vulnerability and Effectiveness of Occupant Protection Technologies for Older Occupants and Women*, DOT HS 811 766, at x (2013).

[4] Jessica Y. Ho, *Life Course Patterns of Prescription Drug Use in the United States*, 60 DEMOGRAPHY 1549, 1561 (2023).

[5] M Rademaker, *Do Women Have More Adverse Drug Reactions?*, 2 AM. J. CLIN. DERMATOL.349, 349 (2001); Nell Tharpe, *Adverse Drug Reactions in Women's Health Care*, 56 J MIDWIFERY WOMENS HEALTH 205, 205 (2011); Irving Zucker and Brian J. Prendergast, *Sex Differences in Pharmacokinetics Predict Adverse Drug Reactions in Women*, 11 BIOLOGY OF SEX DIFFERENCES 1, 12 (2020).

These disparities share a common source: male-default design. Automobile safety systems have historically been engineered and tested using male crash-test dummies, with female bodies treated as variations rather than design inputs.[6] In pharmaceuticals, higher adverse-event rates among women are closely linked to their historical exclusion to longstanding underrepresentation in clinical trials. As a result, drug dosing too often reflects male physiology, despite well-established sex differences in drug absorption and metabolism. In both contexts, products are designed around a male reference model implicitly treated as universal, producing predictable and systematic safety consequences for women.[7]

Male-default design is a structural feature of modern product development, extending across products ranging from safety equipment and tools to medical devices, surgical instruments, and footwear, with increasingly well-documented consequences for women's safety.[8] Automobiles and prescription drugs are not anomalies; they are illustrative of a broader design paradigm in which products optimized for male bodies predictably perform less safely for women, not because women are atypical users, but because they were never incorporated into the design baseline.

Despite these realities, products liability law has never squarely confronted the legal significance of these disparities. Namely, whether a product designed around male bodies, marketed as gender-neutral, and foreseeably more dangerous for women constitutes a design defect.[9] Modern products liability doctrine rests on the premise that products placed into the stream of commerce should be reasonably safe for their intended users.[10] Yet courts have struggled to operationalize that premise where risks are systematically distributed across users. In evaluating design defect claims, courts typically assess product safety in the aggregate—asking whether a design's risks outweigh its benefits across the user population as a whole.[11] Where risks and benefits are unevenly distributed across subgroups of users,

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6 *See infra* Part I.

7 *See infra* Part I.

8 *Id.*

9 Feminist torts scholarship has long demonstrated that ostensibly neutral doctrines of reasonableness, injury, and damages embed male experience as the normative baseline. Yet despite extensive critique of tort law's gendered assumptions, courts and commentators have paid comparatively little attention to how gender bias operates through product design itself. *See, e.g.*, Martha Chamallas & Linda Kerber, *Women, Mothers, and the Law of Fright*, 88 MICH. L. REV. 814 (1990) (examining how negligence doctrines governing emotional distress privilege "masculine" conceptions of injury and discount harms more commonly experience by women, particularly caregivers); Martha Chamallas, *Questioning the Use of Race-Specific and Gender-Specific Economic Data in Tort Litigation*, 63 FORDHAM L. REV. 73 (1994) (critiquing the use of gendered wage data in damages calculations, showing how tort law reproduces labor-market inequality by treating women's lower earnings as objective baselines); Leslie Bender, *A Lawyer's Primer on Feminist Theory and Tort*, 38 J. LEGAL EDUC. 3 (1988) (arguing that tort law's conception of reasonableness assumes male norm); Martha Chamallas, *The Architecture of Bias: Deep Structures in Tort Law*, 146 U. PA. L. REV. 463 (1998) (identifying structures in tort law that systematically privilege male experiences over women experiences). One *exception see Dolly M. Trompeter, Sex, Drugs, and the Restatement (Third) of Torts Section 6(c): Why Comment e is the Answer the Woman Question*, 48 AM. U.L. REV. 1139, (1999) (arguing that this section 6(c) will disproportionately affect women).

10 *Restatement (Third) of Torts: Products Liability* § 2 cmt. a.

11 *See infra* Part III.A.

this aggregate analysis allows a product's strong safety performance for one subgroup—most often men—to mask the fact that a product is systematically unsafe for another—women.[12] The result is a doctrine that risks legitimizing products that embed inequality into their very design.

This Article argues that this outcome reflects a doctrinal blind spot rather than a doctrinal constraint. Existing doctrine does not require courts to conduct an aggregate analysis. Both the risk–utility test and the consumer expectations test—the two leading frameworks for determining design defect—contain conceptual space to evaluate whether a product is defective for a substantial and foreseeable subgroup of ordinary users. Courts have occasionally applied this logic in cases involving children or users with common allergies even when the product is marketed to a much broader population.[13] Yet they have not extended this logic to male-default design cases, even though women constitute a majority of adult consumers and a plainly foreseeable class of users of products marketed as gender neutral.

Building on this insight, the Article advances a new framework for evaluating design defects—one that incorporates subgroup considerations into products liability doctrine and provides the theoretical foundation for finding a male-centric design that is marketed as gender neutral to be defective.[14] It proceeds in two stages. First, it argues that courts should more consistently embrace limited subgroup considerations, explicitly accounting for subgroup outcomes when evaluating defectiveness. At a minimum, products that are marketed as gender neutral should not be insulated from liability merely because superior performance for men masks predictable harms to women. Second, and more ambitiously, the Article proposes the incorporation of robust subgroup considerations into design defect analysis. Under this approach, courts would treat systematic disparities in safety outcomes across men and women as independently relevant to defectiveness. This shift would reorient products liability law from an exclusive focus on aggregate safety toward a framework attentive to how safety risks are allocated among foreseeable users.

This Article makes three contributions to products liability theory and doctrine. First, it identifies male-default design as an analytically distinct and previously untheorized source of design defect within products liability law. While scholars and regulators have documented

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12 *Id.*

13 *See infra* Part III.B.

14 This Article recognizes the conceptual distinction between sex and gender. Sex refers to biological differences, including anatomy and physiology, while gender refers to social roles, identities, and expectations. The safety disparities examined here arise primarily from sex-based biological differences—including anthropometry, biomechanics, and pharmacokinetics—rather than from gender identity or expression. For clarity and consistency with existing products liability doctrine, regulatory language, and case law, the Article at times uses the terms "sex" and "gender" interchangeably, without intending to collapse that distinction or to deny the relevance of gender identity in other legal contexts.

gender bias in product design,[15] this Article is the first to show how products liability doctrine itself facilitates that bias by treating aggregate safety as dispositive and thereby rendering systematic harms to women legally invisible.

Second, the Article develops a doctrinal framework for incorporating subgroup considerations into design defect analysis. Drawing on existing features of products liability law, as well as courts' treatment of subgroup harms in allergy and child-safety cases, the Article demonstrates that recognizing gender-based safety disparities does not require abandoning settled doctrine. Instead, it requires courts to more consistently apply tools they already possess through limited subgroups considerations that accounts for subgroup outcomes. More ambitiously, the Article also contends that the products liability doctrine should adopt a robust approach that treats systematic disparities as independently relevant to defectiveness.

Third, the Article reframes design defect doctrine as a mechanism not only for minimizing aggregate harm, but also for allocating risk in consumer markets. By showing how male-default design allows manufacturers to externalize design costs onto women, the Article clarifies how a subgroup sensitive approach can strengthen deterrence incentives and promote inclusive design, and better align products liability law with its underlying rationales of foreseeability, safety, and fairness.

The Article proceeds as follows. Part I documents the scope and causes of male-default design, beginning with a detailed case study of automobiles and extending to a wide range of other products. Part II introduces the structure of modern products liability law, focusing on design defect doctrine and the concept of the ordinary user. Part III advances the Article's core contribution: a framework for incorporating subgroup considerations into design defect analysis, first through limited attention to subgroup harms and then through a more robust approach that treats systematic disparities as independently relevant to defectiveness. Part IV addresses implementation concerns, including causation and the alternative design requirement associated with the risk-utlity test.

Products liability law has long asked whether a product is sufficiently safe. This Article contends that it must also ask a more basic question: safe for whom? When the answer

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15 Perhaps the most widely acknowledged bias in product design has been prescription drugs. *See* Cynthia Hathaway, *A Patent Extension Proposal to End the Underrepresentation of Women in Clinical Trials and Secure Meaningful Drug Guidance for Women*, 67 FOOD DRUG L.J. 143 (2102); Anna C. Mastroianni, *HIV, Women, and Access to Clinical Trials: Tort Liability and Lessons from DES*, 5 DUKE J. GENDER L. & POL'Y 167 (1998); Shana F. Oppenhein, *Jack & Jill Take Lots of Pills, But Jill Comes Tumbling After: Gender Inequality in Privately Funded Early Phase Clinical Trials*, 22 WM. & MARY J. OF WOMEN & L. 393 (2016).

is "men," and the product is marketed to women as well, that disparity warrants legal scrutiny.

I. PRODUCTS DESIGNED FOR MEN BUT MARKETED TO BOTH SEXES

Modern product markets operate against a largely unexamined design baseline: the male body. Across a wide range of products marketed as gender neutral, male anthropometric and physiological norms continue to structure design, testing, and validation decisions. This reliance on a single reference body is not incidental or product-specific; it reflects a common feature of modern product development. The consequences are predictable. Products optimized around male bodies often perform less safely for women, not because women are atypical users, but because they are treated as deviations from the design norm.

This Part begins by offering a detailed case study of automobiles—a product category where sex-based design disparities have been especially well documented and have life-or-death consequences. It then extends the analysis across other product domains to demonstrate that male-default design is a common feature of modern production rather than an isolated design failure. The Part concludes by examining the institutional and economic forces that sustain this baseline and, in doing so, systematically shape whose safety is prioritized in consumer markets.

A. AUTOMOBILES: A UBIQUITOUS PRODUCT DESIGNED FOR MEN

Automobile crashes provide a particularly clear lens through which to examine male-default design. Driving is a routine activity for most adults, and given the broad and heterogeneous population of drivers, one might expect vehicle safety systems to protect users equally. Yet crash data consistently show that women experience significantly worse outcomes than men in otherwise comparable collisions. Female drivers are substantially more likely to suffer serious or moderate injury and face a higher risk of death, even after controlling for factors such as seatbelt use and crash severity.[16] These disparities make automobiles a paradigmatic case in which products marketed to all users perform differently—and less safely—for women.

1. *How Did We Get Here?*

The National Highway Traffic Safety Administration (NHTSA) is charged with reducing injuries, deaths, and economic losses from car crashes.[17] It pursues this mandate primarily through two mechanisms. First, NHTSA sets the federal minimum performance

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[16] *See* Bose, Segui-Gomez & Crandall, s*upra* note 3, at 2371; Miner et al., *supra* note 3, at 8; Kahane, *supra* note 3, at x.

[17] Title II of Pub. L. 91–605, 84 Stat. 1713, enacted Dec. 31, 1970, at 84 Stat. 1739.

requirements for new motor vehicles and equipment, known as the Federal Motor Vehicle Safety Standards (FMVSS).[18] Some standards specify crash avoidance requirements, such as those related to brakes, while others specify crashworthiness requirements, such as seatbelts or airbags.[19] Second, NHTSA administers the New Car Assessment Program (NCAP), a consumer information program that rates vehicle safety on a five-star scale based on comparative crash performance.[20]

Both FMVSS compliance and NCAP ratings rely heavily on crash test dummies as proxies for human occupants.[21] These dummies are instrumented with sensors at critical anatomical locations—including the head, neck, chest, and abdomen—that record acceleration, force, and related measures during controlled collisions.[22] Researchers use these data, often in combination with cadaver studies, to generate injury risk curves estimating the probability of specific injuries at varying force levels.[23] Regulators then translate these curves into injury criteria—numerical thresholds intended to represent the limits of human tolerance to forces such as head impact, chest compression, or neck loading.[24] Vehicles that exceed these thresholds fail FMVSS requirements and cannot be sold in the United States,[25] while NCAP converts the same data into relative risk scores that determine a vehicle's star rating.[26]

Since the 1970s, NHTSA has mandated the type of crash test dummies that must be used in tests for FMVSS compliance and NCAP ratings, thereby standardizing the metrics through which vehicle safety is evaluated.[27] Early NHTSA crash test dummies were modeled

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18 The FMVSS are located in 49 C.F.R. § 571. Manufacturers must certify that their motor vehicles or motor vehicle equipment comply with applicable FMVSS before their motor vehicles or motor vehicle equipment can be sold in the U.S. See 49 U.S.C. §§ 30112, 30115.

19 49 C.F.R § 571.208.

20 *See* Nat'l Highway Traffic Safety Admin., DOT HS 810 698, The New Car Assessment Program Suggested Approaches for Future Program Enhancements at 3-4 (2007), *available at* http://www.safercar.gov/staticfiles/DOT/ safercar/pdf/810698.pdf; Lawrence L. Hershman, National Highway Traffic Safety Administration United States Paper Number 390 at 3, *The U.S. New Car Assessment Program (NCAP): Past, Present and Future*, available at https://www-nrd.nhtsa.dot.gov/pdf/esv/esv17/Proceed/00245.pdf (noting that "Compared to the 48.3 km/h (30 mph) FMVSS No. 208 compliance tests, the 8 km/h (5 mph) faster NCAP crash tests produce a 36-percent increase in crash energy.").

21 U.S. Gov't Accountability Off., GAO-23-105595, Vehicle Safety DOT Should Take Additional Actions to Improve the Information Obtained from Crash Test Dummies 7 (2023).

22 *Id.*

23 *Id.*

24 *Id.*

25 Michael Kleinberger, et al., *Development of Improved Injury Criteria for the Assessment of Advanced Automotive Restraint Systems*, NHTSA 12-13 (Sept. 1998), *available at* chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.nhtsa.gov/sites/nhtsa.gov/files/criteria.pdf.

26 Each vehicle receives summary scores, representing the relative risk of injury for occupants in different seating positions and types of crash tests. These scores are then combined and converted into the vehicle's star rating. Generally, the lower the estimated probability of injury, the higher the vehicle rating. Lawrence L. Hershman, National Highway Traffic Safety Administration United States Paper Number 390, *The U.S. New Car Assessment Program (NCAP): Past, Present and Future*, at 4 available at https://www-nrd.nhtsa.dot.gov/pdf/esv/esv17/Proceed/00245.pdf.

27 NHTSA does so through incorporating the dummy into its regulations using the same rulemaking process it uses to issue new or amended FMVSS. The purpose of specifying design requirements for the dummy types is to ensure consistency in

on military prototypes designed for pilot ejection-seat testing[28] and were, predictably, modeled on the 50th-percentile male.[29] In 1978, NHTSA mandated the use of what would become the most widely used crash test dummy in the world: the Hybrid III 50th-percentile male dummy.[30] Designed in 1976, the Hybrid III represent a man who is 5 feet 9 inches tall and weighs 173 pounds—approximately the average U.S. male in the 1970s.[31] Its design embodies male muscle-mass distribution, joint structures, and spinal column.[32]

Although NHTSA has expanded its testing protocols to reflect a broader range of crash scenarios such as side-impact and rollover accidents—its reliance on the 50th-percentile male dummy has remained remarkably static.[33] Notably, the Hybrid III 50th-percentile male dummy continues serve as the primary model for evaluating safety measures during vehicle testing.[34] The consequences of this regulatory history are significant. Safety system such as seatbelts and airbags calibrated using the 50th-percentile Hybrid III male dummy, undeniable perform better for bodies that approximate its dimensions. As a result, women who are on average smaller, lighter, and have different biomechanical vulnerability have been systematically disadvantaged.

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the crash testing ratings. Testing conditions for NCAP crash tests are outlined in NHTSA's periodic updates to NCAP, which are published in the Federal Register.

28 Marek Jaskiewicz et al., *Overview and Analysis of Dummies Used for Crash Tests*, 35 SCIENTIFIC JOURNALS 107, 22 (2013).

29 *Id.*

30 449 C.F.R. Part 572, Subpart E;

31 49 C.F.R. Part 572, Subpart E; *see also* Marek Jaskiewicz et al., *Overview and Analysis of Dummies Used for Crash Tests*, 35 SCIENTIFIC JOURNALS 107, 23 (2013). The Hybrid III 50% male dummy also features six high-strength steel ribs that can simulate human chest deflection and lower torso with a rubber lumbar spine that curves like ours when seated. *Id.*

32 Marek Jaskiewicz et al., *Overview and Analysis of Dummies Used for Crash Tests*, 35 SCIENTIFIC JOURNALS 107, 23 (2013).

33 Lawrence L. Hershman, National Highway Traffic Safety Administration United States Paper Number 390, *The U.S. New Car Assessment Program (NCAP): Past, Present and Future*, at 2 available at https://www-nrd.nhtsa.dot.gov/pdf/esv/esv17/Proceed/00245.pdf (noting that side-impact crash tests were added in 1997 and roll over testing in 2001). At least the side-impact crash tests were included because Congress enacted statutes directing NHTSA to adopt a wide variety of safety standards in specific areas, including side-impact. *See* Jerry L. Mashaw & David L. Harfst, *From Command and Control to Collaboration and Deference: The Transformation of Auto Safety Regulation*, 34 YALE J. ON REG. 167, 183 (2017).

34 In the early 1980s, researchers at the University of Michigan proposed the inclusion of a small female, large male, and average female dummy to better represent the diversity of the driving public. Keith Barry, The Crash Test Bias: How Male-Focused Testing Puts Female Drivers at Risk, CR Consumer Reports (Oct. 23, 2019), *available at* https://www.consumerreports.org/car-safety/crash-test-bias-how-male-focused-testing-puts-female-drivers-at-risk/. NHTSA initially supported the proposal but reversed course during the Reagan Administration, which pursued an aggressive deregulatory agenda aimed in part at easing burdens on the automobile industry. MARISSA M. GOLDEN, WHAT MOTIVATES BUREAUCRATS? POLITICS AND ADMINISTRATION DURING THE REAGAN YEARS 43 (2000) (describing how the Reagan Administration targeted the NHTSA for deregulation). The 1980s brought budget cuts that reduced the NHTSA's resources, a departure of roughly one-third of its professional staff, and the abandonment of the plan to incorporate an average female dummy. NHTSA's budget was reduced from $259 million in 1979 to $211 million in 1988 and its budget for research and analysis was cut by one-third. MARISSA M. GOLDEN, WHAT MOTIVATES BUREAUCRATS? POLITICS AND ADMINISTRATION DURING THE REAGAN YEARS 44 (2000); Mashaw & Harfst, *supra* note 33, at 192. One factor that likely contributes to the NHTSA slow rate of adopting new crash test dummies is legal culture of the agency. Jerry Mashaw and David Harfst document that from 1986 to 2002 NHTSA effectively stopped rulemaking in response to a host of factors, including leadership by new political appointees with little enthusiasm for the Agency's mission. *Id.* at 182–83 (calling 1987-2002 the "Ice Age of Rulemaking" for NHTSA and noting that "significant rulemaking at NHTSA atrophied nearly to the point of extinction" during this period).

2. *Automobiles Disproportionately Harm Women*

Studies consistently demonstrate that women face greater risks of injury and death in automobile crashes than men. Women are 26% more likely to suffer chest injuries, 45% more likely to sustain neck injuries, 58% more likely to experience arm injuries, 80% more likely to sustain leg injuries, and 17% more likely to die in comparable collisions.[35] These disparities are not random. They are closely linked to vehicle designs calibrated around the average male body, which leave women disproportionately vulnerable.

Women differ from men in ways that are directly relevant to crash safety. On average, women have less upper-body muscle mass, shorter sitting heights, lower body weight, wider pelvic measurements, and shorter torsos.[36] These differences affect how female bodies interact with car seats, head restraints, airbags, and seatbelts during collisions, altering how crash forces are transmitted through the body and increasing injury risk.

Whiplash injuries illustrate this dynamic.[37] Women experience whiplash at significantly higher rates than men, with some studies indicating that women are up to three times more likely to suffer whiplash in rear-end crashes.[38] This elevated risk reflects both physiological differences—such as a higher head-to-neck ratio, smaller neck circumference, and lower neck strength—and design choices that fail to accommodate those differences.[39] Seats optimized for average-sized male occupants tend to propel lighter occupants forward more rapidly, while overly rigid seatbacks absorb energy less effectively for smaller bodies.[40] Head restraints are also frequently misaligned for women; because of shorter sitting heights, women's heads often contact the lower portion of the restraint, increasing neck injury risk rather than mitigating it.[41]

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[35] Kahane, *supra* note 3.

[36] Anita Berglund et al., *Occupant- and Crash-Related Factors Associated With the Risk of Whiplash Injury*, 13 ANN EPIDEMIOL. 66, 70 (2003).

[37] The economic burden of whiplash injury is substantial, amounting to nearly $4 billion dollars annually in the United States in medical care, disability, and lost productivity. Jason C. Eck, Scott D. Hodges & S. Craig Humphreys, *Whiplash: A Review of a Commonly Misunderstood Injury*, 110 AM. J. MED. 651, 651 (2001).

[38] A.P. Morris & P. Thomas, *Neck Injuries in the UK Co-Operative Crash Injury Study*, SAE Technical Paper 962433 at 317 (1996); Bertil Jonsson et al., *The Risk of Whiplash-Induced Medical Impairment in Read-End Impacts for Males and Females in Driver Seat Compared to Front Passenger Seat*, 37 IATSS RESEARCH 8, 10 (2013); Brian O'Neill et al., *Automobile Head Restraints—Frequency of Neck Injury Claims in Relation to the Presence of Head Restraints*, 62 AM. J. PUBLIC HEALTH 399, 403 (2012).

[39] Bertil Jonsson et al., *The Risk of Whiplash-Induced Medical Impairment in Read-End Impacts for Males and Females in Driver Seat Compared to Front Passenger Seat*, 37 IATSS RESEARCH 8, 10 (2013).

[40] David C. Viano, *Seat Influences on Female Neck Responses in Rear Crashes: A Reason Why Women Have Higher Whiplash Rates*, 4 TRAFFIC INJURY PREVENTION 228, 228 (2010).

[41] Janella F. Chapline et al., *Neck Pain and Head Restraint Position Relative to the Driver's Head in Rear-End Collisions*, 32 ACCIDENT ANALYSIS AND PREVENTION 287, 292 (2000) (noting that "females with adequately position . . . head restraints were significantly less likely to report neck pain . . than females with poorly position head restraints"). *Id.* at 287 (noting that "females with adequately position . . . head restraints were significantly less likely to report neck pain . . than females with poorly position head restraints").

Similar design dynamics contribute to other injury disparities. Women's shorter stature often places them in "out-of-position" seating postures, as they sit further forward and more upright to reach pedals and maintain visibility.[42] This positioning increases vulnerability to internal injuries in frontal crashes.[43] Early-generation airbags amplified this risk: calibrated for average-sized men and deploying with excessive force, they disproportionately injured women, who were more likely to be seated closer to the steering wheel.[44] A NHTSA investigation of early airbag fatalities found that more than 75% of driver deaths involved women.[45]

Women also suffer higher rates of lower-extremity injuries in crashes.[46] Differences in bone density, leg length, and foot size alter lower-body kinematics during impact, while poor seatbelt fit further reduces protection.[47] Women's wider pelvic dimensions and shorter stature affect how seatbelts distribute force across the body, diminishing their effectiveness and increasing injury risk.[48]

3. *Updates in Regulations are Not Enough*

In 2003, the NHTSA mandated the first "female" dummy for crash testing.[49] Yet, the female dummy adopted by NHTSA—the Hybrid III female dummy—does not represent the average woman and is female only in name. To begin, the Hybrid III female dummy is only 4 feet 9 inches tall and weighs a mere 101 pounds, representing less than 5% of women based on mid-1970s standards.[50]

Second, the female dummy fails to reflect the physiological differences between women and men. Rather than being designed on the basis of female anatomy, it is simply a

[42] CAROLINE CRIADO PEREZ, INVISIBLE WOMEN: DATA BIAS IN A WORLD DESIGNED BY MEN at 186 (2019).
[43] Kahane, *supra* note 3.
[44] NHTSA, SPECIAL CRASH INVESTIGATIONS: FIRST GENERATION FRONTAL AIR BAGS A MODEL FOR FUTURE CORRECTIVE ACTION, DOT HS 811 261, at 3 (Jan. 2010), available at https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/811261.
[45] David C. Viano, *History of Airbag Safety Benefits and Risks*, 25 TRAFFIC INJURY PREVENTION 268, 268 (2024). NHTSA promulgate a rule requiring automakers to install advanced front airbag systems for the driver and front passenger. Each system must include sensors to detect the size of the occupant, the seat position, seatbelt use and the severity of the crash and then deploy the airbag with less force if necessary.
[46] Kahane, *supra* note 3, at 124.
[47] *Id.*; Patricia C. Dischinger, Timothy J. Kerns & Joseph A. Kufera, *Lower Extremity Fractures in Motor Vehicle Collisions: The Role of Driver Gender and Height*, 27 ACCIDENT ANALYSIS & PREVENTION 601, 605 (2003).
[48] John P. Forman et al., *Occupant Anthropometry and Belt Fit: A Study of Driver Belt Positioning*, 57 Stapp Car Crash Journal 1, 1–24 (2013) (showing belt fit varies substantially with stature—a factor that contribute to sex-based disparities in belt effectiveness); H. Saito, B. Pipkorn & N. Lubbe, *Understanding the Influence of Seat Belt Geometries on Belt-to-Pelvis Angle*, SAE Tech. Paper No. 2022-01-0873 (2022) (using human body models to show female and small-stature occupants have higher risk of improper belt-to-pelvis engagement and submarining).
[49] GAO, *supra* note 21, at Fig. 3.
[50] *Id.* at Fig. 6; Humanetics, Hybrid III 5th Female, https://www.humaneticsgroup.com/products/anthropomorphic-test-devices/frontal-impact/hybrid-iii-5th-female; 49 C.F.R. Part 572 Subpart-0; 65 Fed. Reg. 10968 (Mar. 1, 2000), as amended by 67 Fed. Reg. 46415 (July 15, 2002).

scaled-down version of the Hybrid III 50th-percentile male dummy.[51] This ignores critical distinctions—differences in muscle mass distribution, bone density, vertebrae spacing, center of gravity, and pelvic structure—that researchers believe help explain women's elevated crash injury rates.

Third, the female dummy is not consistently placed in the driver's seat for all front- and side-impact crash tests.[52] For FMVSS compliance tests, the dummy occupies the driver's position in two of the three tests.[53] For NCAP ratings, she appears in only one of three.[54] This limited use of the 5th-percentile female dummy in the crash tests diminishes its potential to improve safety outcomes for women drivers.

Finally, injury risk curves and injury criteria used to interpret dummy data are based overwhelmingly on male physiology.[55] Cadaver research has predominantly focused on male subjects that resemble the 50th-percentile male dummy, meaning that female risk curves are typically just scaled-down versions of male curves.[56] Although some criteria are adjusted for sex, the absence of female-specific data hampers the ability to identify and correct crash risks unique to women.

Despite its limitations, the 5th-percentile female dummy has spurred some improvements in safety features for women and smaller occupants, such as refinements to air bags and seatbelts. Although the differences in injury risks between males and females have narrowed with newer model vehicles, studies still consistently confirm that women remain at greater risk of injury than men.[57] As long as the "average man" remains the design baseline, women will continue to experience disproportionate harms on the road.

There is some indication that this paradigm may begin to shift. In November 2025, the NHTSA released the specifications for the THOR-05F dummy, a new advanced female crash test dummy that better mimics female anatomy.[58] The release of its design details is

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51 GAO, *supra* note 21, at 23 ("According to NHTSA, the Hybrid III 5th-percentile female dummy is essentially a scaled-down version of the 50th-percentile male dummy.").

52 *Id.* at 26.

53 *Id.*

54 *Id.*

55 *Id.*at 28–29.

56 *Id.*

57 NHTSA, SEX-BASED DIFFERENCES IN ODDS OF MOTOR VEHICLE CRASH INJURY OUTCOMES, (Jan. 2026), *available at* https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/813754 (assessing injury odds in females versus males in variety of injury outcomes within six crash types in nationally representative set of data over 22 years and finding that females have higher odds of injury than males in 39 out of 150 models and for the remaining 111 models, females had non-significantly higher injury odds in 81); NHTSA, FEMALE CRASH FATALITY RISK RELATIVE TO MALE FOR SIMILAR PHYSICAL IMPACTS tbl.4 at 22 (Aug. 2022), *available at* https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/813358 (estimating that female front-row occupants have a 19.9% greater risk of death than males when in vehicles from model years 1960 to 1999 and a 9.4% greater risk of death than males when in vehicles from model years 2000 to 2020).

58 U.S. Dep't of Transportation, *Trump's Transportation Secretary Sean P. Duffy Unveils Design for First-of-its-Kind Advanced Female Crash Test Dummy*, Nov. 20, 2025, https://www.transportation.gov/briefing-room/trumps-transportation-secretary-sean-p-duffy-unveils-design-first-its-kind-advanced.

a crucial step towards its adoption, although it is not yet mandated for either NCAP or FMVSS crash tests.[59]

B. Male-Default Design as a Systemic Feature of Modern Products

Automobiles are not the only products designed around male bodies, marketed to both sexes, and disproportionately harmful to women. Across consumer and professional domains, manufacturers repeatedly rely on male anthropometric and physiological norms as design baselines. Although these products differ in function and context, the resulting harms follow a small number of recurring design mechanisms. Grouping these examples by mechanism—rather than by product—reveals the systematic nature of male-default design and avoids treating each instance as an isolated anomaly.

1. *Hand Size, Grip Strength, and Force Requirements*

Many products are designed around assumptions about hand size, grip span, and upper-body strength that reflect male averages. Hand tools such as drills, hammers, and screwdrivers are typically built for men's larger hands and greater grip strength. On average, women's hands are approximately one inch shorter than men's,[60] and the median female grip strength is only about 64% of the male median.[61] Fewer than 0.1% of women exceed the median male grip strength.[62]

These differences have direct safety implications. Widely spaced handles require women with average or small hands to overstretch to maintain control, while tools requiring sustained or peak force place repetitive strain on weaker musculature. As a result, women experience faster fatigue, reduced control, and higher rates of carpal tunnel syndrome and other repetitive stress injuries. In occupations that require frequent tool use, women are more likely than men to develop musculoskeletal disorders—not because they are untrained users, but because the tools are designed around a different physiological baseline.[63]

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59 *Id.*

60 Jenna Fletcher, *What is the Average Hand Size?*, Medical News Today (2023), *available at* https://www.medicalnewstoday.com/articles/average-hand-size#glove-size.

61 Frank M. Perna, Kisha Coa, Richard P. Troiano, et. Al, *Muscular Grip Strength estimates of the U.S. Population from the National Health and Nutrition Examination Survey 2011–2012*, 30 J. Strength Con Res. 867, 872 (2016).

62 *Id.*

63 *Id.*

This same logic persists in surgical instruments.[64] Although women now comprise roughly half of U.S. medical school applicants[65] and nearly one quarter of surgical residents,[66] surgical tools remain largely designed for male hands.[67] As one surgeon observed, contemporary instruments are often too large and require excessive force, sometimes stretching women's hands to their limits or requiring two-handed operation.[68] These mismatches contribute to markedly higher rates of upper-limb musculoskeletal disorders among female surgeons.[69] Studies report that 88% of female orthopedic surgeons experience ergonomic injuries, compared to 45% of their male counterparts,[70] with similar disparities documented among laparoscopic surgeons.[71]

2. Reach, Stature, and Control Accessibility

A second recurring mechanism involves assumptions about height, reach, and body strength embedded in the design of equipment with fixed controls or operator cabins. Heavy machinery—including tractors, forklifts, cranes, and backhoes—is typically designed around the average male operator's stature and reach.[72] For women, this often results in reduced visibility, controls that are difficult or impossible to reach, and seating configurations that force awkward or unsafe postures.[73]

These design choices increase injury risk in multiple ways. Straining to reach pedals or steering mechanisms can compromise control and situational awareness while also producing long-term musculoskeletal strain. Empirical research shows that women working in industries reliant on heavy machinery face higher rates of musculoskeletal injuries such as strains, sprains, and hernias, as well as elevated risks of acute injuries including

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64 Anna Braman & Christopher Robertson, *Should Current Laws Be Revised to Address Occupational Hazards Caused by Hand-Tool Size Mismatch Among Surgeons?*, 159 JAMA SURGERY 359, 359 (2024) (noting that surgical tools need to either be designed with principles of universal design to make tools "work just as well for larger and small hands" or "tools need to be designed in multiple sizes").

65 Association of American Medical Colleges, 2024 Facts: Applicants and Matriculants Data, A-1 U.S. Medical School Applications and Matriculants by School, State of Legal Residence, and Gender, 2024-2025 *available at* https://www.aamc.org/data-reports/students-residents/data/facts-applicants-and-matriculants.

66 R. Berguer & A. Hreljac, *The Relationship Between Hand Size and Difficulty Using Surgical Instruments: A Survey of 726 Laparoscopic Surgeons,* 18 SURG. ENDOSC. 508, 508 (2004).

67 Young Eun Koo, Charlotte Allen, Angela Ballantyne, et al., *Androcentric Bias in Surgical Equipment – What Challenges do Women Face?*, 227 AM. J. SURGERY 106,106 (2024) ("Most surgical tools were developed at a time when the surgical workforce was predominantly and sometimes exclusively male, leading to an androcentric (male focus) bias in product design.").

68 Jim Rudnick & Jessica Liberatore, *Addressing Women's Needs in Surgical Instrument Design*, Medical Device and Diagnostic Industry (2006), *available at* https://www.mddionline.com/surgical/addressing-women-s-needs-in-surgical-instrument-design.

69 Berguer & Hreljac, *supra* note 66, at 508 (finding "[h]and size is a significant determinant of the difficulty using laparoscopic surgical instruments").

70 Brianna Fram, Meghan E. Bishop, Pedro Beredjiklian, & Daniel Seigerman, *Female Sex is Associated with Increased Reported Injury Rates and Difficulties with Use of Orthopedic Surgical Instruments*, 13 CUREUS 14952, at 6 (2021).

71 Berguer & Hreljac, *supra* note 66, at 509.

72 Oyebode A Taiwo, Linda F Cantley, Martin D. Slade, et al., *Sex Differences in Injury Patterns Among Workers in Heavy Manufacturing*, 169 AM. J. EPIDEMIOL. 161, 164 (2008).

73 *Id.*

lacerations and contusions.[74] Again, these injuries arise not from misuse or inexperience, but from foreseeable interactions between women's bodies and equipment calibrated to male dimensions.

3. Protective Equipment and Fit-Dependent Safety

Male-default assumptions also shape the design of personal protective equipment (PPE), where proper fit is a prerequisite for effective protection. Women comprise the vast majority of nurses and a large share of health care workers,[75] yet PPE has historically been designed using male body data derived from mid-twentieth-century populations.[76] Gloves, gowns, masks, and eye protection frequently fail to fit women properly, undermining their protective function.[77]

Ill-fitting PPE increases exposure to physical, chemical, and biological hazards.[78] Oversized gloves reduce dexterity and tool control, while improperly fitting eye protection increases vulnerability to splashes and debris.[79] PPE designs also often fail to account for women's biological realities, including pregnancy and menopause, leading to overheating, discomfort, and unsafe workarounds.[80] Where protection depends on fit, male-default sizing predictably compromises safety for women.[81]

4. Dynamic Load Transfer and Injury Amplification: Footwear Design

A fourth mechanism involves products that shape how forces move through the body during motion. Athletic footwear provides a clear example. For decades, women's athletic shoes were largely produced by scaling down male prototypes,[82] ignoring well-documented

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74 *Id.*

75 Boniol M, McIsaac M, Xu L, Wuliji T, Diallo K, Campbell J., *Gender Equity in the Health Workforce: Analysis of 104 Countries. Geneva: World Health Organization* (2019), available at https://apps.who.int/iris/handle/10665/311314.

76 D.J. Janson, B.G. Clift, V. Dhokia, *PPE Fit of Healthcare Workers During the COVID-19 Pandemic*, 99 APPLIED ERGONOMICS 103610, 103610 (2021).

77 Women are four times more likely to report their surgical gowns as being large to some degree than men and twice as likely to experience oversized surgical masks than men. D.J. Janson, B.G. Clift, V. Dhokia, *PPE Fit of Healthcare Workers During the COVID-19 Pandemic*, 99 APPLIED ERGONOMICS 103610, 10360 (2021) (finding that 55% of women and 13% of men reported their surgical gowns being too large and 54% of women and 27% of men reported their surgical masks as being oversized); *see also* Hignett, R. Welsh R, & J. Banerjee, *Human Factors Issues of Working in Personal Protective Equipment During the COVID-19 Pandemic*, 76 ANAESTHESIA 134 (2020).

78 During the COVID-19 pandemic, female health care workers were more likely than their male colleagues to become infected, a disparity linked in part to poor equipment fit. K. El-Boghdadly, D.J.N. Wong, R. Owen, et al., *Risks to Healthcare Workers Following Tracheal Intubation of Patients with COVID-19: A Prospective International Multicentre Cohort Study*, 75 ANAESTHESIA 1437, 1444 (2020).

79 S. Hignett, R. Welsh R, & J. Banerjee, *Human Factors Issues of Working in Personal Protective Equipment During the COVID-19 Pandemic*, 76 ANAESTHESIA 134 (2020).

80 Women in Global Health, Fit for Women? Safe and Decent PPE for Women Health and Care Workers, at 40 (Nov. 2021), available at chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://gendro.org/wp-content/uploads/2023/11/1_WGH-Fit-for-Women-report-2021.pdf.

81 *Id.* at 45.

82 Yuyu Xia et al., *Gender Differences in Footwear Characteristics Between Half and Full Marathons in China: A Cross-Sectional Survey*, 13 SCIENTIFIC REPORTS 13020, 13021 (2023); Adriana Wehrens, Women's-Specific Soccer Boots are Here to

anatomical differences in foot structure. Women tend to have proportionally higher arches, wider forefeet, and narrower heels than men—differences that are biomechanically significant.[83]

Footwear that fails to accommodate these structural variations alters lower-limb mechanics. Inadequate arch support and excessive heel movement destabilize the kinetic chain from the ankle through the knee, increasing joint torque and stress during cutting, landing, and pivoting movements. These altered biomechanics contribute directly to elevated injury risk. Female athletes are two to ten times more likely than men to suffer anterior cruciate ligament (ACL) tears, a disparity that cannot be fully explained by neuromuscular or training differences alone.[84] While multiple factors contribute to ACL injury risk, footwear design that assumes a male foot morphology amplifies biomechanical stresses for women.[85]

5. Sex-Specific Physiology and Biomedical Design

A final mechanism involves products whose safety depends on physiological processes that differ systematically by sex. Prescription drugs provide a clear example. For decades, women were excluded from early-phase clinical trials, and even when included, their data were rarely analyzed separately.[86] Although women's participation in clinical trials has increased substantially since the mid-1990s—following changes in federal research policy and FDA guidance—meaningful gaps remain, particularly in certain therapeutic areas.[87] As a result, dosing guidelines and safety profiles have historically been calibrated primarily to male physiology, despite well-established sex differences in drug absorption, distribution, metabolism, and excretion.[88]

The consequences of this design choice are well documented. Women experience higher rates of adverse drug reactions, in part because the standard practice of prescribing

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Change the Game, Urban Pitch (March 24, 2023) *available at https://urbanpitch.com/womens-specific-soccer-boots-are-here-to-change-the-game/*.

83 Roshna E. Wunderlich & Peter R. Cavanagh, *Gender Differences in Adult Foot Shape: Implications for Shoe Design*, 33 MED. SCI. SPORTS EXERC. 605 (2001).

84 Editorial, *The Female ACL: Why is it More Prone to Injury*, 13 J. ORTHOPAEDICS AI (2016).

85 *Id.*; Molly Longman, *What if More Women Designed Running Shoes?*, Refinery29 (March 26, 2020), *available at* https://www.refinery29.com/en-us/women-vs-men-running-shoe-lasts-feet-difference; Craig Welch, *Why Are So Many Teen Girls Still Tearing Their A.C.L.s?*, N.Y.TIMES (Feb. 20, 2026), *available at* https://www.nytimes.com/2026/02/26/magazine/acl-tear-women-girl-sports.html.

86 INSTITUTE OF MEDICINE, WOMEN AND HEALTH RESEARCH 41 (1994) (noting that 1977 the FDA issued guidelines that women of childbearing years should be excluded from early stage clinical trials). The FDA reversed course in 1993 and issued new guidelines on the analysis of clinical trial data by gender. Guideline for the Study and Evaluation of Gender Differences in the Clinical Evaluation of Drugs, 58 FED. REG. 39406–39416. The new guideline revises the section titled "Women of Childbearing Potential" in the FDA's 1977 guidelines, General Considerations for the Clinical Evaluation of Drugs, and describes the agency's expectations regarding inclusion of both genders as subjects of clinical trials of drugs. Clinical trial data is often not adequately analyzed by gender. *Id.* at 48.

87 Karen M. Goldstein, et al., *Strategies for Enhancing the Representation of Women in Clinical Trials: An Evidence Map*, 13 SYSTEMATIC REV. 2, 2 (2024).

88 Heather Whitley & Lindsey Wesley, *Sex-Based Differences in Drug Activity*, 80 AM. FAMILY PHYSICIAN 1254 (2009); OP Soldin & DR Mattison, *Sex Differences in Pharmacokinetics and Pharmacodynamics*, 48 CLIN PHARMACOKINET 143 (2009);

identical doses to women and men ignores pharmacokinetic differences.[89] The FDA's 2013 decision to halve the recommended dose of Ambien for women—after evidence showed dangerously elevated blood levels at the prior "standard" dose—illustrates how male-centered design produces foreseeable harms that often emerge only after widespread use.[90] Pharmaceutical design thus reflects the same baseline problem as physical products: a male reference model treated as universal, with predictable safety consequences for women.

C. WHY ARE WOMEN MISSING FROM DESIGN?

Women's underrepresentation in product design persists for structural reasons that bear directly on products liability law. Product design and engineering have long been male-dominated fields, shaping professional norms, reference models, and testing practices.[91] As a result, male anthropometric and physiological data became embedded early as default design baselines, influencing standards and practices that continue to govern product development today.

Economic incentives have reinforced this default. Designing products around a single reference body allows manufacturers to streamline research and development, minimize the expense of collecting sex-disaggregated data, and avoid the costs associated with developing multiple prototypes or size ranges. Even when evidence demonstrates that women's anthropometric and ergonomic needs diverge significantly from men's, efficiency considerations often favor treating variation as deviation rather than as a design constraint. These incentives help explain why male-default design persists even when female users are foreseeable and numerically significant.

Regulatory structures have compounded this dynamic. As the automobile case study illustrates, agencies tasked with setting safety standards have often failed to displace male-centered baselines, even in the face of mounting evidence of sex-based injury disparities. NHTSA's decades-long reliance on male crash-test dummies, and its slow adoption of more representative female models, reflects broader institutional pressures identified by Jerry

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89 M Rademaker, *Do Women Have More Adverse Drug Reactions?*, 2 AM. J. CLIN. DERMATOL.349, 349 (2001); Nell Tharpe, Adverse Drug Reactions in Women's Health Care, 56 J MIDWIFERY WOMENS HEALTH 205, 205 (2011); Irving Zucker and Brian J. Prendergast, *Sex Differences in Pharmacokinetics Predict Adverse Drug Reactions in Women*, 11 BIOLOGY OF SEX DIFFERENCES 1, 12 (2020).

90 CBS News, FDA: Cut Ambien Dosage for Women, Jan. 10, 2013, *available at* https://abcnews.go.com/Health/fda-recommends-slashing-sleeping-pill-dosage-half-women/story?id=18182165. Although a later study suggested the lower dosing may not be supported by data. *See* David J. Greenblatt, Jerold S. Harmatz & Thomas Roth, *Zolpidem and Gender: Are Women Really at Risk?*, 39 J. CLINICAL PSYCHOPHARMACOLOGY 189 (2019).

91 AMY SUE BIX, GIRLS COMING TO TECH!: A HISTORY OF AMERICAN ENGINEERING EDUCATION FOR WOMEN, MIT Press 1–2 (2014) (noting that well into the twentieth century, leading engineering programs and professional societies openly excluded women and that women comprised less than one percent of engineering students nationwide in the 1950s).

Mashaw and David Harfst: industry resistance, judicial scrutiny, and political constraints that discourage forward-looking, expert-driven regulation.[92] Over time, NHTSA retreated from ambitious standard-setting and instead relied on recall authority to address discrete defects, leaving systemic design disparities—such as male-centered testing protocols—largely uncorrected.[93]

Market responses have not filled this gap. When manufacturers have sought to expand into women's markets, they have frequently done so without investing in inclusive research or redesign.[94] Instead, products are frequently scaled down from male-oriented designs or cosmetically altered—a practice colloquially described as "shrink it and pink it."[95] Such strategies preserve the appearance of gender neutrality while perpetuating male-centered design choices.

Finally, even well-intentioned designers may default to male standards because of the persistent data gaps. Many industries lack systematic information on how women interact with products, how female physiology affects safety performance, or how design features operate across different body types.[96] Where such data are absent, women's needs are effectively invisible in the research, testing, and prototyping stages.[97] This gender data gap allows male-centered designs to replicate themselves across generations of products, while simultaneously limiting the availability of evidence needed to identify safer alternatives.[98]

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In sum, this Part demonstrates that male-default design extends far beyond automobiles and recurs across a wide range of products integral to daily life and professional practice. Where products are designed around male anthropometric and physiological norms but marketed to a broad population, women encounter predictable safety disadvantages. These harms are not the result of idiosyncratic vulnerability or misuse, but arise from

92 JERRY L. MASHAW & DAVID L. HARFST, THE STRUGGLE FOR AUTO SAFETY (1990)
93 *Id.*
94 Cf. A similar issue arises in recruiting Black patients into clinical trials where it is estimate that recruiting Black respondents is significantly more expensive than white respondents, contributing to the underrepresentation of Black patients in clinical trials. Marcella Alsan, Maya Durvasula, Harsha Gupta, Joshua Schwartzstein & Hedi Williams, *Representation and Extrapolation: Evidence from Clinical Trials*, 139 Q.J.ECON. 575, 586 (2024).
95 Karen Korellis Reuther, Shrink It and Pink It: Gender Bias in Product Design, Harvard Advanced Leadership Initiative Social Impact Review, available at https://www.sir.advancedleadership.harvard.edu/articles/shrink-it-and-pink-it-gender-bias-product-design.
96 CRIADO PEREZ, *supra* note 42, at XV–XIX.
97 *Id.*
98 *Id.*

ordinary interactions between women's bodies and products calibrated to a different reference model.

These disparities could, in principle, be addressed through legislative or regulatory intervention. In practice, however, regulation alone is ill suited to the scope and structure of the problem. Statutory mandates have the advantage of establishing baseline protections ex ante—before injuries occur—rather than relying on after-the-fact compensation. By requiring inclusive testing and sex-disaggregated data prior to market entry, statutes and regulations could close persistent gender data gaps and ensure that safety standards reflect the diversity of product users. Moreover, statutory rules create clarity and predictability for manufacturers, reducing the uncertainty of case-by-case adjudication and ensuring uniform national standards. However, broad statutory and regulatory interventions present their own difficulties.

Statutes and regulations generally operate at a high level of generality, establish minimum floors rather than optimal design targets, and cannot anticipate every design choice that allocates risk among users. Meaningful regulatory reform would require agencies to resolve difficult calibration questions and to update standards as scientific knowledge evolves—tasks that have proven challenging in practice.[99] In a regulatory environment marked by institutional constraint and retrenchment, reliance on regulation alone is unlikely to eliminate male-default design.

Moreover, male-default design is not confined to a narrow set of highly regulated products, but characterizes much of the modern consumer marketplace, including tools, safety equipment, and consumer goods for which no centralized regulator comparable to the FDA exists. Even in areas subject to extensive regulation, male-centered baselines have persisted despite decades of evidence of disproportionate harms. The drawbacks of statutes and regulations suggest a residual role for tort law.

As a result, products liability law remains a critical site for evaluating designs that perform well for some users while systematically imposing greater risks on others.[100] While imperfect, litigation can generate market-based incentives for manufacturers to design more inclusively and can evolve incrementally as judicial awareness of gender disparities grows. Accordingly, the remainder of this Article explores how products liability doctrine might be leveraged to achieve these aims—specifically, how courts can incorporate subgroup

[99] *See* Wendy Wagner, *When All Else Fails: Regulating Risky Products Through Tort Litigation*, 95 GEO. L.J. 693, 695 (2007).
[100] *See* Mary J. Davis, *The Products Liability Restatement: Was It a Success?: On Restating Products Liability Preemption*, 74 BROOKLYN L. REV. 759, 776–79 (2009) (arguing for a residual role for the products liability litigation considering limitations of regulation).

considerations into products liability doctrine to recognize as legally defective those products designed for men, marketed to all, and foreseeably injurious to women.

II. PRODUCTS LIABILITY LAW: DESIGN DEFECTS

This Part sets the stage for the subgroup analysis deveoped in Part III by introducing the structure of modern products liability law. It outlines the two principal tests used to evaluate design defectiveness—the risk–utility test and the consumer expectations test—and explains how each frames the assessment of product safety and design reasonableness. The Part then examines the concept of the "ordinary user," which serves as the reference point for defect determinations under both tests.

A. PRODUCTS LIABILITY LAW

Products liability law is an area of law that holds manufacturers, distributors, suppliers, and retailers responsible when a defective product injures a consumer. Unlike ordinary negligence, product liability is often founded on strict liability, meaning a manufacturer can be liable for a defective product regardless of intent or carefulness.[101] Over the past century, the law evolved from requiring contractual privity and proof of negligence to imposing strict tort liability for consumer product injuries, ensuring that the costs of such injuries are borne by the manufacturers rather than the injured consumers.[102]

There are three different categories of products liability claims: manufacturing defects, failure to warn, and design defects.[103] Manufacturing defects are flaws arising from the production process.[104] A manufacturing defect exists when a product departs from its intended design due to some error in fabrication, assembly, or materials.[105] Because the product deviated from the manufacturer's specifications, these cases are often

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101 *Restatement (Second) of Torts* § 402A.

102 Douglas A. Kysar, *The Expectations of Consumers*, 103 COLUM. L. REV. 1700, 1709-10 (2003); Gary T Schwartz, *New Products, Old Products, Evolving law, Retroactive Law*, 58 N.Y.U. L. REV. 796, 797–811 (1983); WARD FARNSWORTH AND MARK F. GRADY, TORTS CASES AND QUESTIONS 385–400 (3rd ed. 2019); *But see*, Alexandra D. Lahav, *A Revisionist History of Products Liability*, 122 MICH. L. REV. 509, 519–554 (2023) (arguing that the privity of contract did not take hold in American courts but instead negligence era dominated much longer than believed).

103 *See, e.g.*, *Restatement (Second) of Torts* § 402A; Shanks v. Upjohn Co., 835 P.2d 1189, 1194 (Alaska 1992); Ulrich v. Kasco Abrasives Co., 532 S.W.2d 197, 200 (Ky. 1976); Voss v. Black & Decker Mfg. Co., 450 N.E.2d 204, 207 (N.Y. 1983).

104 *See, e.g.*, Restatement (Third) of Torts §2, comment a. (1997); N.J. Stat. Ann. § 2A:58C-2 (West 1987) (noting that a manufacturer is liable if a claimant proves that the product "deviated from the design specifications, formulae, or performance standards of the manufacturer or from otherwise identical units manufactured to the same manufacturing specifications or formulae"); Barker v. Lull Eng'g Co., 573 P.2d 443, 454 (Cal. 1978) ("[A] defective product is one that differs from the manufacturer's intended result or from other ostensibly identical units of the same product line."); *Rix v. General Motors Corp.*, 723 P.2d 195, 200 (Mont. 1986) (noting that a product is defective if the manufacturer did not construct it correctly and if it does not conform to its intended design).

105 *Id.*

straightforward: if the product was malformed and caused harm, it is considered defective under strict liability.[106]

Failure to warn claims involve failures to adequately warn or instruct users about latent dangers of the product.[107] A product may be perfectly manufactured and well-designed but still pose certain risks that are not obvious to the average consumer. In such cases, the manufacturer has a duty to provide sufficient warnings or instructions so that consumers can use the product safely. In warning-defect cases, liability often hinges on whether the risk was known or knowable and whether the warning given was adequate for an ordinary user.[108]

Finally, there are design defects.[109] In a design defect, the product is manufactured as intended, but the intended design itself is dangerous. The focus is on the decisions made in the design process such as the materials chosen, the dimensions or safety mechanisms, rather than on factory errors. A classic definition is that a design defect exists when a product is designed in a way that poses an unreasonable risk of injury to consumers, especially when a safer, economically and technologically feasible alternative design could have been used.[110]

B. DESIGN DEFECTS

The Article's focus is on the last type of products liability claims—design defect, as the primary concern of this Article is firms designing products for too narrow of a user range. Importantly, firms are marketing the product as gender neutral but making design choices that are male-centric. As a result, the product either does not work as well for women or worse disproportionately harms them. There are two primary tests to judge the adequacy of a product's design: (1) the consumer expectations test and (2) the risk-utility test. Although products liability claims are strict liability offenses, as explained below, these tests also sound in negligence.

1. *The Consumer Expectations Test*

Comment i to section 402A of the *Restatement (Second) of Torts* first articulated the modern consumer expectations test.[111] Under this test, a product is defective if it is "in a condition not contemplated by the ultimate consumer, which will be unreasonably dangerous

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106 *Id.*
107 *See, e.g.*, *Restatement (Third) of Torts* §2(c).
108 *See, e.g.*, Anderson v. Owens-Corning Fiberglas Corp., 810 P.2d 549, 557 (Cal. 1991) (en banc) (stating that actual or constructive knowledge of risk is a requisite for failure-to-warn action); Fibreboard Corp. v. Fenton, 845 P.2d 1168, 1175 (Colo. 1993) (en banc) (same); Olson v. Prosoco, Inc., 522 N.W.2d 284, 289 (Iowa 1994) (plaintiff "must prove a defendant knew or should have known of potential risks associated with the use of its product, yet failed to provide adequate directions or warnings to users").
109 *See, e.g.*, *Restatement (Third) of Torts* §2(b) & §1, comment a.
110 *Restatement (Third) of Torts: Prods. Liab*. § 2(b).
111 *Restatement (Second) of Torts* § 402A (1965).

to him."[112] A product is unreasonably dangerous if it poses risks "beyond that which would be contemplated by the ordinary consumer who purchases it, with the ordinary knowledge common to the community as to its characteristics."[113] Thus, a plaintiff prevails by showing that the product failed to meet the safety standards an ordinary, reasonable consumer would expect when using it in a reasonably foreseeable manner.[114] Whether a particular product violated these expectations is a question of fact for the jury to decide.[115]

Almost immediately, commentators noted serious problems with this approach.[116] Most significantly, they argued that consumer expectations are too amorphous to serve as a workable standard for assessing design defectiveness.[117] Factfinders, they claimed, need more concrete guidance than the vague benchmark of consumer expectations.[118] This concern is particularly acute for technically complex products, where consumers often lack specific expectations regarding critical features. For example, while a consumer may understand that automobiles are not crashproof, they are unlikely to have meaningful expectations about the degree to which the passenger compartment will remain intact in a side-impact collision

112 *Restatement (Second) of Torts* § 402A cmt. i.

113 *Id.*

114 Douse v. Bos. Sci. Corp., 314 F. Supp. 3d 1251, 1251 (M.D. Fla. 2018); Show v. Ford Motor Co., 697 F. Supp. 2d 975 980–81 (N.D. Ill. 2010).

115 Gutterman v. Target Corp., 242 F. Supp. 3d 695, 706 (N.D. Ill. 2017) ("Typically, application of the consumer-expectation test is a task for the jury, but it can be decided as a matter of law where no reasonable jury could find that a product performed other than how an ordinary consumer would expect." (citation omitted).

Additionally, courts disfavor finding for a plaintiff when the risk was open and obvious, *e.g.*, Delaney v. Deere & Co., 999 P.2d 930, 935-39, 946 (Kan. 2000) (rejecting "open and obvious" danger test with respect to design defect claims). Some courts have held that an open and obvious risk cannot violate consumer expectations because any reasonable consumer would know that such a risk exists, *e.g.*, Austin v. Clark Equip. Co., 48 F.3d 833, 836 (4th Cir. 1995). Other courts have maintained flexibility within the consumer expectations test by holding that the obviousness of a danger is simply a factor for courts to consider when determining the expectations of a consumer, *Delaney*, 999 P.2d at 946. Along a similar vein, courts are hesitant to find that products which establish consumer expectations through adequate warnings violated consumer expectations unless the warning itself was inadequate, *see* Thongchoom v. Graco Children's Prods., Inc., 71 P.3d 214, 218–19 (Wash Ct. App. 2003).

116 James A. Henderson, Jr. & Aaron D. Twerski, *Achieving Consensus*, 83 CORNELL L. REV. 867, 882 (1998). For further criticism of the doctrine, *see* Michael D. Bernachhi, *A Behavioral Model for Imposing Strict Liability in Tort: The Importance of Analyzing Product Performance in Relation to Consumer Expectation and Frustration*, 47 U. CIN. L. REV. 43, 46 (1978) ("the apparent widespread adoption of [the consumer expectations] standard is form almost void of substance . . ."); Jane Stapleton, *Restatement (Third) of Torts: Products Liability, an Anglo-Australian Perspective*, 39 WASHBURN L.J. 363, 378 (2000) ("In a sense, the consumer expectations test simply gives the fact-finder its head. . . . It nakedly invites the fact-finder to use his or her gut instincts to determine the dispute.'").

117 W. PAGE KEETON ET AL, PROSSER AND KEETON ON THE LAW OF TORTS § 99, at 699 (5th ed. 1984) ("The meaning is ambiguous and the test is very difficult of application to discrete problems. . . . [As a result, t]he test can be utilized to explain most any result that a court or jury chooses to reach."); Sheila L. Birnbaum, *Unmasking the Test for Design Defect: From Negligence [to Warranty] to Strict Liability to Negligence*, 33 VAND. L. REV. 593, 611-18 (1980) (warning that consumer expectations test might lead to "confused, divergent, and unjust decisions"); James A Henderson, Jr., *Symposium, A Discussion and a Defense of the Restatement (third) of Torts: Products Liability*, 8 KAN J.L. & PUB. POL'Y 19, 20 (1998) ("If you look at the scholarship prior to 1992 . . . major players in this products liability field almost without exception . . . said consumer expectations will not work as a mainstream test."); Donald T. Ramsey, *Consumer Expectation Test for Design Defect in California*, 24 TORT & INS. L.J. 650, 668 (1984); *see also* John H. Chun, *The New Citadel: A Reasonably Designed Products Liability Restatement*, 79 CORNELL L. REV. 1654, 1673–74 (1994).

118 This sentiment has led at least one commentator to conclude that the consumer expectations test "is so vague as to be lawless." Henderson & Twerski, supra note 116, at 882.

at fifty miles per hour.[119] Unsurprisingly, several states have concluded that the consumer expectation test is inappropriate for evaluating the design of complex products.[120]

Scholars have also leveled narrower criticisms. Some argue the consumer expectations test effectively a negligence analysis incognito. Because a "reasonable consumer" would only expect cost-effective safety features, the test collapses into a negligence framework without its formal structure. Indeed, many jurisdictions purporting to apply the consumer expectations test have in practice incorporated negligence principles directly into the analysis.[121] Other commentators warn that the test may actually inhibit innovation in safety features. Because consumer expectations are shaped by the existing state of product design, they often lag behind technological advances. As a result, manufacturers may avoid liability even when inexpensive safety improvements are available.[122]

The controversy surrounding the consumer expectations test eventually prompted the American Law Institute (ALI) to adopt the *Restatement (Third) of Torts* in 1997. The *Restatement (Third) of Torts: Products Liability*[123] rejected the consumer expectations test as an independent test for design defect, replacing it with the analytically rigorous risk-utility test.[124] In doing so, the ALI aligned itself with scholarly consensus favoring an explicit cost-benefit balancing approach to determining the defectiveness of a product.

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119 Kysar, *supra* note 102, at 1716; John E. Montgomery & David G. Owen, *Reflections on the Theory and Administration of Strict Tort Liability for Defective Products*, 27 S.C. L. REV. 803, 823 (1976).

120 For example, court applying the law of Alaska, *Gen. Motors Corp. v. Farnsworth*, 965 P.2d 1209, 1221 (Alaska 1998), Florida, *Tran v. Toyota Motor Corp.*, 420 F.3d 1310, 1314 (11th Cir. 2005), and Arizona, *Brethauer v. GMC*, 221 ARIZ. 192, 211 P.3d 1176, 1184 (Ct. App. 2010), have all found that the design of seat belts may be tested by the consumer expectation standard. Nevertheless, other states explicitly allow the consumers expectations test for complex products. See, e.g., *Soule v. General Motors Corp.*, 882 P.2d 298, 309–310 (Cal. 1994); Tincher v. Omega Flex, 628 Pa. 296,

121 Other critiques include that to the extent consumer attitudes reflect only a generalized hope that the product will not hurt them, then the consumer expectations test threatens to morphs into absolute manufacturer liability, a standard that no American jurisdiction has accepted. Kysar, *supra* note 102, at 1708 ("Despite frequent academic support for the establishment of . . . 'enterprise liability,' American courts invariably have stopped short of making the manufacturer bear the personal injury costs of all product-caused accidents irrespective of fault."). Additionally, although the vagueness and amorphousness of the test has been argued to favor the plaintiff, s*ee, e.g.*, Philip H. Corboy, *The Not-So-Quiet Revolution: Rebuilding Barriers to Jury Trial in the Proposed Restatement (Third) of Torts: Products Liability*, 61 TENN. L. REV. 1043, 1089 (1994) (noting practical difficulties for plaintiffs in absence of consumer expectations test); John F. Vargo, *Caveat Emptor: Will the A.L.I. Erode Strict Liability in the Restatement (Third) for Products Liability?*, 10 TOURO L. REV. 21, 33 (1993) (suggesting courts should be more sympathetic to the "consumer's viewpoint"), other commentators have objected to the consumer expectations test because the doctrine can too easily work against the plaintiff, Kysar, *supra* note 102, at 1716.

122 *See* Reed Dickerson, *How Good Does a Product Have to Be?*, 42 IND. L.J. 301, 314 (1967) ("[C]onsumer expectations will lag behind actual practices, and [therefore] the further extension of safety practices designed to minimize the consequences of accidents will depend either on direct regulation or on the development of a broader rationale of seller responsibility."); David A. Fischer, *Products Liability—The Meaning of Defect*, 39 MO. L. REV. 339, 349–50 (1974) (noting that consumer expectations "will not always be in line with what the reasonable manufacturer can achieve" and therefore manufacturers "will[, at times,] have no incentive to improve the safety of products").

123 *Restatement (Third) of Torts: Prods. Liab.*.

124 The new *Restatement* rests design defect liability on a risk-utility balancing approach in which costs and benefits of a proposed safety improvement to the product design are weighed in order to determine whether the absence of the improvement renders the marketed design "not reasonably safe." *Id.* § 2(b).

Nevertheless, courts have not uniformly embraced the *Third Restatement's* demotion of the consumer expectations test. Fifteen states still employ the consumer expectation test exclusively to design defect claims,[125] while others adopt a hybrid approach, allowing either the consumer expectations test, risk-utility test, or both.[126]

2. *Risk-Utility Test*

The risk-utility test has emerged as the dominant framework for evaluating design defectiveness.[127] The articulation of this test depends upon whether the jurisdiction requires proof of an alternative reasonable design. In jurisdiction that do, a product "is defective in design when the foreseeable risks of harm posed by the product could have been reduced or avoided by the adoption of a reasonable alternative design," and the "omission of the alternative design renders the product not reasonably safe" for the ordinary user.[128] In jurisdictions that do not require proof of a reasonable alternative design, the inquiry is simply whether the utility of the product's design justifies the risks it creates.[129] Unlike the

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125 Arkansas, Indiana, Kansas, Maryland, Nebraska, New Hampshire, North Dakota, Oklahoma, Oregon, Rhode Island, Tennessee, Utah, Vermont, Wisconsin, and Wyoming exclusively apply the consumer expectations test. ARK. CODE ANN. § 16-116-102(7)(A) (2016); IND.CODE ANN. § 34-20-4-1 (West 2011); North Dakota, N.D. CENT. CODE § 28-01.3- 01(4) (2016); Tennessee, TENN. CODE ANN. § 29-28-102(8) (2012); Brown v. Sears, Roebuck & Co., 328 F.3d 1274, 1278–79 (10th Cir. 2003) (discussing Utah's consumer expectation's test); Austin v. Lincoln Equip. Assocs., 888 F.2d 934, 936 (1st Cir. 1989); Delaney v. Deere & Co., 999 P.2d 930, 945–47 (Kan. 2000); Halliday v. Sturm, Ruger & Co., 792 A.2d 1145, 1152 (Md. 2002); Freeman v. Hoffman-La Roche, Inc., 618 N.W.2d 827, 834 (Neb. 2000); Vautour v. Body Masters Sports Indus., Inc., 784 A.2d 1178, 1181 (N.H. 2001); Oklahoma, Woods v. Fruehauf Trailer Corp., 765 P.2d 770, 774–76 (Okla. 1988); McCathern v. Toyota Motor Corp., 23 P.3d 320, 331 n.15 (Or. 2001); Zaleskie v. Joyce, 333 A.2d 110, 113–14 (Vt. 1975); Green v. Smith & Nephew AHP, Inc., 629 N.W.2d 727, 739–41 (Wis. 2001); Sims v. General Motors Corp., 751 P.2d 357, 364–65 (Wyo. 1988); *see also* Mike McWilliams & Margaret Smith, *An Overview of the Legal Standard Regarding Product Li8ability Design Defect Claims and a Fifty State Survey on the Applicable Law in Each Jurisdiction*, 82 DEF. COUNS. J. 80, 83–85 (2015).

126 Alaska, Arizona, California, Connecticut, Florida, Hawaii, Illinois, Mississippi, and Washington recognize both tests. GMC v. Farnsworth, 965 P.2d 1209, 1220 (Alaska 1998); Arizona, Golonka v. GMC, 65 P.3d 956, 962–63 (Ariz. Ct. App. 2003); Soule v. GMC, 882 P.2d 298, 308–09 (Cal. 1994); Potter v. Chi. Pneumatic Tool Co., 694 A.2d 1319, 1330 (Conn. 1997); Force v. Ford Motor Co., 879 So. 2d 103, 106 (Fla. Dist. Ct. App. 2004); Acoba v. General Tire, Inc., 986 P.2d 288, 304 (Haw. 1999); Calles v. Scripto-Tokai Corp., 864 N.E.2d 249, 255 (Ill. 2007); Glenn v. Overhead Door Corp., 935 So. 2d 1074, 1081 (Miss. Ct. App. 2006); Soproni v. Polygon Apartment Partners, 971 P.2d 500, 504–05 (Wash. 1999); *see also* Mike McWilliams & Margaret Smith, *An Overview of the Legal Standard Regarding Product Li8ability Design Defect Claims and a Fifty State Survey on the Applicable Law in Each Jurisdiction*, 82 DEF. COUNS. J. 80, 87–88 (2015).

127 Alabama Colorado, Georgia, Idaho, Kentucky, Louisiana, Massachusetts, Michigan, Minnesota, New Jersey, New Mexico, New York, North Carolina, Ohio, Pennsylvania, South Carolina, Texas, and West Virginia exclusively apply the risk-utility test.; Louisiana, LA. STAT. ANN. § 9:2800.54 (2018); North Carolina, N.C. GEN. STAT. ANN. §§ 99B-6(A), (B), 99B11 (West 2011); Ohio, OHIO REV. CODE ANN. § 2307.75 (LexisNexis 2017); Pennsylvania, Surace v. Caterpillar, Inc., 111 F.3d 1039, 1042 (3rd Cir. 1997) (discussing Pennsylvania's risk-utility test); Minnesota, Ehlers v. Siemens Medical Solutions, USA, Inc., 251 F.R.D. 378, 383–384 (D. Minn. 2008); Flemister v. GMC, 723 So.2d 25, 27–28 (Ala. 1998); Camacho v. Honda Motor Co., 741 P.2d 1240, 1246–47 (Colo. 1987); Jones v. NordicTrack, Inc., 550 S.E.2d 101, 103–104 (Ga. 2001); Toner v. Lederle Labs., 732 P.2d 297, 306 (Idaho 1987); Kentucky, Ostendorf v. Clark Equip. Co., 122 S.W.3d 530, 535 (Ky. 2003); Evans v. Lorillard Tobacco Co., 990 N.E.2d 997, 1013–1014 (Mass. 2013); Gregory v. Cincinnati Inc., 538 N.W.2d 325, 333 (Mich. 1995); Jurado v. W. Gear Works, 619 A.2d 1312, 1317–18 (N.J. 1993); Brooks v. Beech Aircraft Corp., 902 P.2d 54, 62 (N.M. 1995); Scarangella v. Thomas Built Buses, Inc., 717 N.E.2d 679, 681–682 (N.Y. 1999); Branham v. Ford Motor Co., 701 S.E.2d 5, 14 (S.C. 2010); Uniroyal Goodrich Tire Co. v. Martinez, 977 S.W.2d 328, 335 (Tex. 1998); Beatty v. Ford Motor Co., 574 S.E.2d 803 (W. Va. 2002); see also McWilliams & Smith, supra note 4, at 85–87 (collecting references)

128 *Restatement (Third) of Torts: Prods. Liab*. § 2(b).

129 *See infra* note 184.

consumer expectations test, the focus is not on perceptions of consumers but on the reasoned decision-making of the manufacturer.

Although no universally accepted list of factors governs the risk-utility analysis, courts most frequently consider the following seven: (1) the usefulness of the product; (2) the likelihood the product will cause injury and the seriousness of that injury; (3) the availability of a safer alternative product; (4) the manufacturer's ability to make the product safer without impairing the utility of the product; (5) the user's ability to avoid danger in the product; (6) the degree of awareness of the potential danger of the product which reasonably can be attributed to the plaintiff; and (7) the ability of the manufacturer to acquire insurance or incorporate losses into the price of the good.[130] If, after weighing these factors, the factfinder determines that the risks of the product's design outweigh its benefits, the product is deemed unreasonably dangerous. Where risks and benefits differ across users, the factfinder typically compares the aggregate risks to the aggregate benefits.[131]

Although products liability law is generally considered a strict liability offense, design defect doctrine incorporates negligence principles.[132] The test requires manufacturers to adopt safer designs only when safety benefits outweigh the costs, including reductions in usefulness and increases in price and production expenses. While the *Restatement* does not explicitly equate the risk-utility test with negligence, courts have consistently recognized the parallels.[133]

The test has also not escaped scholarly criticism. Scholars argue that the requirement of a reasonable alternative design —adopted in certain jurisdictions—imposes too heavy a burden on plaintiffs, particularly in cases involving technically complex products.[134] Demonstrating an alternative design can be prohibitively expensive, thereby frustrating one of the original aims of strict liability: alleviating the plaintiff's evidentiary disadvantage of the lack access to the information needed to prove negligence.[135] Others fault the test for

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[130] These factors were originally proposed by John W. Wade. John W. Wade, *On the Nature of Strict Liability*, 44 MISS. L.J. 825, 837–38 (1973) (describing the "Wade" factors for the risk-utility test); *see also* Aaron D. Twerski & James A. Henderson, *Manufacturers' Liability for Defective Product Designs: The Triumph of the Risk-Utility*, 74 BROOKLYN L. REV. 1061, 1080–89 (2009) (reviewing the use of Wade factors by each court that uses risk-utility test for design defects).

[131] James A. Henderson, Jr. & Aaron D. Twerski, *Drugs Designs Are Different*, 111 YALE L.J. 152, 168–170 (2001) (discussing how aggregation of benefits and risks is a necessary compromise in determining the defectiveness for nonprescription drug products).

[132] James A. Henderson, Jr. & Aaron D. Twerski, *Doctrinal Collapse in Products Liability: The Empty Shell of Failure to Warn*, 65 N.Y.U. L. REV. 265, 271-272 (1990) ("[C]ost-benefit balancing is also at the heart of negligence.")

[133] *See* David G. Owen& Mary J. Davis, 1 Owen & Davis on Products Liability § 5:6, at 412 ("Many courts have pointed to the functional equivalence of strict liability and negligence in . . . design cases.").

[134] John F. Vargo, *The Emperor's New Clothes: The American Law Institute Adorns a "New Cloth" for Section 402A Products Liability Design Defects—A Survey of the States Reveals a Different Weave*, 26 U. MEMPHIS L. REV. 493, 517 (1996); Frank J. Vandall, *The Restatement (Third) of Torts: Product Liability Section 2(b): The Reasonable Alternative Design Requirement*, 61 TENN. L. REV. 1407, 1423 (1994).

[135] *See, e.g.*, Cronin v. J.B.E. Olson Corp., 401 P.2d 1153, 1162 (Cal. 1972) ("[T]he very purpose of . . . [the development of strict products liability] was to relieve the plaintiff from problems of proof inherent in pursing negligence. . . .").

failing to provide guidance on which factors are the most important or explain how courts should weigh tradeoffs.[136] For instance, when a product offers significant utility but also carries a substantial risk of serious injury, the framework provides no clear standard for resolving the tradeoff. Despite these criticisms, the risk-utility test remains the dominant and generally preferred method for judging design defectiveness.

3. The Ordinary User

In both the risk-utility test and the consumer expectations test, the benchmark for determining whether a product is defective is the ordinary user or ordinary consumer. Under the consumer expectation test, a product is unreasonably dangerous if it is "dangerous to an extent beyond that which would be contemplated by the ordinary consumer who purchases it . . . ."[137] By contrast, under the risk-utility test a product is not reasonable safe if the utility of the product does not justify its risks for the ordinary user in jurisdictions that do not require proof of a reasonable alternative design. In jurisdictions that do require such proof, a product "is defective in design when the foreseeable risks of harm posed by the product could have been reduced or avoided by the adoption of a reasonable alternative design," and the "omission of the alternative design renders the product not reasonably safe" for the ordinary user.[138]

Who, then, is the ordinary user? The ordinary user is not an identifiable individual but a hypothetical average person expected to use the product in a normal and foreseeable manner.[139] The concept resembles the reasonable person standard in negligence law, which serves as the benchmark for assessing whether conduct is negligent.[140] Unlike the reasonable person, however, the ordinary user operates within a market context in which products are designed for and marketed to particular groups of consumers. As a result, manufacturers can—and should—anticipate the characteristics of their intended users. Courts are therefore more willing to account for the attributes of the hypothetical "ordinary user" than those of the abstract "reasonable person."[141] Accordingly, courts have considered factors such as

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136 *See, e.g.*, W. KIP VISCUSI, REFORMING PRODUCTS LIABILITY 72–73 (1991).

137 *Id.*

138 *Restatement (Third) of Torts: Prods. Liab.* § 2(b).

139 Some courts, however, have reasoned it is not appropriate to compensate a plaintiff who was aware of the risk that a product presented. See, e.g., Magnuson v. Rupp Mfg., Inc., 171 N.W.2d 201, 203–05 (Minn. 1969).

140 Christopher Brett Jaeger, *The Empirical Reasonable Person*, 72 ALA. L. REV. 887, 897 (2021) ("[T]he objective reasonable person standard is entrenched in American tort law."). .

141 The reasonable person standard is generally not tailored to the characteristics of a particular defendant, though narrow exceptions exist. For example, a child's conduct is judged against that of a reasonable child of similar age, *Restatement (Third) of Torts: Liab. For Physical & Emotional Harm* § 10 ("A child's conduct is negligent if it does not conform to that of a reasonably careful person of the same age, intelligence, and experience [subject to two exceptions]."); Bragan ex rel. Bragan v. Symanzik, 687 N.W. 2d 881, 884–85 (Mich. Ct. App. 2004), and physical impairments are factored into a reasonable person standard, *Restatement (Third) of Torts: Liab. For Physical & Emotional Harm* § 11 (a) ("The conduct of an actor with a physical disability is negligent only if the conduct does not conform to that of a reasonably careful person with the same disability."); Kerr v. Connecticut Co., 140 A. 751, 752 (Conn. 1928) (holding a deaf person is held to the standard of the reasonable deaf person). By contrast, factors such as intelligence, *Vaughan v. Menlove*, (1837) 132 Eng. Rep. 490 (holding that diminished

age,[142] gender,[143] and skill level[144] in identifying the relevant ordinary user when products are marketed to a subgroup of the population.

This Article is not concerned with products that are marketed primarily to discrete subgroups, such as products that are marketed exclusively to women, in which courts generally understand the ordinary consumer to be female. Instead, the Article focuses on products that are marketed to a broad population—i.e., as gender neutral. In these cases, as Part III explains, courts typically focus only on the average user rather than discrete subgroups.

III. DESIGN DEFECT DOCTRINE THAT INCORPORATES SUBGROUP CONSIDERATIONS

This Part argues that courts should treat products as defectively designed under products liability law when those products, though marketed to both sexes, are designed around male users in ways that disproportionately endanger women. To that end, it develops a doctrinal framework for integrating robust subgroup considerations into design defect analysis.[145]

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intelligence is not factored into the reasonable person standard), gender, and advanced age, Roberts v. Ring, 173 N.W. 437, 438 (Minn. 1919) (declining to take into account old age as a factor of the reasonable person standard), are typically excluded from the negligence inquiry

142 In *Lugo by Lopez v. LJN Toys*, the plaintiff was an infant injured when another child detached the shield from a Voltron toy figurine and threw it at her, 75 N.Y.2d 850 (1990). The figurine, which was marketed to children four and older, replicated the television character's practice of hurling his shield at enemies, *Id.* at 852. The plaintiff alleged the figurine was defective because the shield was detachable, *Id.* The court treated the relevant ordinary consumer not as an adult but as subgroup which the products was marketed—a child as young as four, *id.*

143 Courts also take gender into account when products are explicitly designed and marketed for one sex. In *Shahbaz v. Johnson & Johnson*, the plaintiff alleged that a transvaginal mesh product was defectively designed due to excessive rigidity and contraction, which caused pain and complications, 2020 WL 5894590 (C.D. Cali. 2020). Because the device was designed exclusively for female reproductive health, the ordinary consumer was properly understood to be a woman, *id.*

144 In *Lamer v. McKee Industries*, 721 P.2d 611 (Alaska 1986), a garage door repairman was killed when a garage door spring—also known as a winding plug—broke and caused him to fall, *Id.* at 612. The court held that the ordinary consumer was the professional garage door repairman and because event experts in the trade did not expect winding plug failures, the spring was deemed defective, *Id.* at 614. Or consider *Gaines-Tabb v. ICI Explosives*, 160 F.3d 613 (10th Cir. 1998) victims of the Oklahoma City bombing sued the manufacturer of ammonium nitrate fertilizer, alleging it was defectively designed because it could be used to make explosives, *Id.* at 624. The plaintiffs argued that the manufacturer could have made the ammonium nitrate safer by incorporating additives to prevent the ammonium nitrate from detonating. *Id.* The court rejected the claim, reasoning that the ordinary consumer of fertilizer was a farmer, and there was no indication that the ammonium nitrate was less safe than expected by farmers. *Id.*

145 The focus of this Article has been on products that disproportionately harm women rather than products that disproportionately benefit men. The distinction between benefits and costs, however, is not always clear. Cars arguably serve as an example of a product that differentially harms women; they provide roughly similar benefits to both sexes but, because they are designed for the average male, disproportionately harm females. Voice recognition systems, which were initially trained on male data are an example of a product that differentially benefits men. Studies have demonstrated that these systems recognize women's voices with less accuracy than men's, thus implying that at least some voice recognition software offer more benefits to men than women. Additionally, depending on how these voice recognition systems are utilized, they may also disproportionately harm women.

The analysis proceeds in three stages. Section A describes products liability doctrine as it currently stands, in which subgroup considerations are typically ignored and superior performance for men can obscure predictable harms to women. Section B advances the Article's central claim: a new framework for evaluating design defects—one that incorporates subgroup considerations into products liability doctrine and provides the theoretical foundation for finding a male-centric design that is marketed as gender neutral to be defective. Section C builds on the foundation in Section B by examining how subgroup considerations may also be incorporated in the reasonable alternative design of the risk-utility test.

A. NO SUBGROUP CONSIDERATIONS

The dominant approach to design defect analysis in American products liability law for a product that is marketed to a large population is to evaluate product safety in the aggregate which obfuscates subgroup considerations.[146] Under this framework, courts determine whether a product is defectively designed by weighing its risks and benefits to the overall population, without disaggregating outcomes across subgroups of users—i.e., women and men. This aggregation reflects the baseline practice in modern products liability doctrine which often assumes that a single, population-wide measure of risk and utility suffices to capture the product's safety profile.

Such an approach, however, risks obscuring important disparities across subgroups of users. Aggregate analysis can mask the fact that a product imposes disproportionate harms on certain subgroups—particularly when the product is designed around a prototypical or "average" user who does not represent the full range of consumers in practice.

To illustrate this conceptual point, consider the following example. A car seat is designed to mitigate whiplash injuries from rear-end collisions. The manufacturer, however, relied exclusively on a male crash-test dummy in its design process. As a result, women experience significantly higher rates of whiplash than men, and the risk–utility profile of the product varies systematically across subgroups.

1. Risk-Utility Test

Under the risk–utility test, courts weigh the risks of harm posed by a product's design against its benefits, often considering the feasibility and cost of safer alternatives. For purposes of illustration, this subsection assumes a jurisdiction in which proof of a reasonable

[146] James A. Henderson, Jr. and Aaron D. Twerski, *Drug Design Liability: Farewell to Comment k*, 67 BAYLOR L. REV. 521, 533 (2015) (noting that for "most products courts adopt an aggregative-welfare perspective" in products liability cases); James A. Henderson, Jr. & Aaron D. Twerski, *Drugs Designs Are Different*, 111 YALE L.J. 152, 168–170 (2001) (discussing how aggregation of benefits and risks is a necessary compromise in determining the defectiveness for nonprescription drug products).

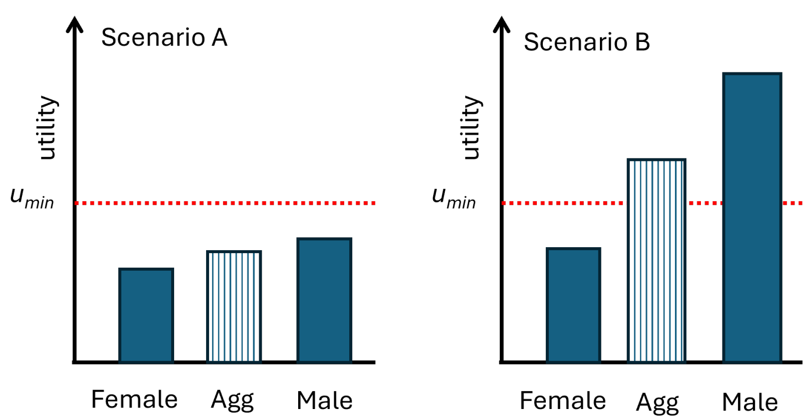
alternative design is not required and the inquiry turns solely on whether the design's utility justifies its risks.

Assume that the benefits and risks associated with the design of a product can be represented by one measure of utility, u. Figure 1 compares the utility of a standard car seat design for female users, male users, and the aggregate population. The design is defective if utility, u, falls below a minimum threshold, u_{min}. An analysis that ignores subgroup considerations evaluates only the aggregate utility, which reflects the net benefits and costs of the product across all users.

In Scenario A, the aggregate utility of the car design—the blue stripped box—falls below the minimum utility value. The design is therefore defective under the risk–utility test. Because the utility for both female and male users also falls below the threshold, the omission of subgroup considerations has no consequence on the outcome.

In Scenario B, by contrast, the aggregate utility exceeds minimum utility value, so the design is not deemed defective. This result holds even though the utility for female users falls below the minimum threshold. When courts consider only aggregate utility, the higher utility enjoyed by male users offsets the elevated risk to women, and the product is treated as reasonably safe despite being defective for a substantial subgroup of users. In this way, a subgroup blind risk–utility analysis can obscure subgroup harm by allowing superior performance for men to outweigh serious risks to women.

Figure 1:



2. The Consumer Expectations Test

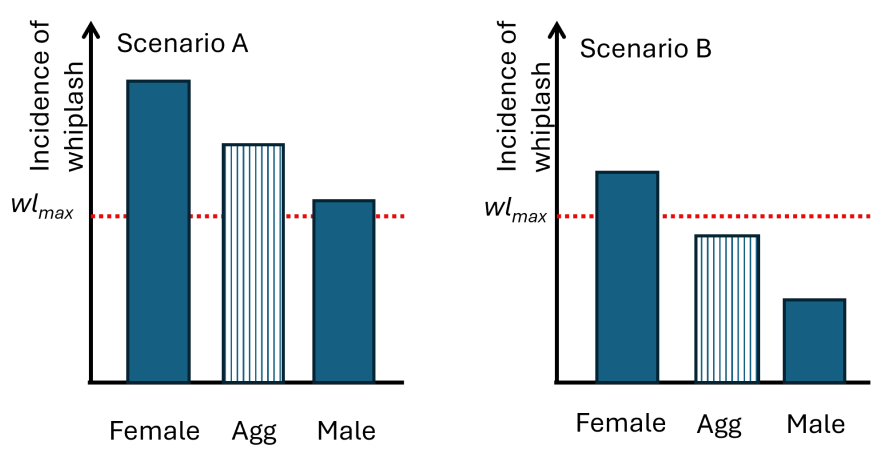
The consumer expectations test raises a distinct, though related, problem. Rather than balancing costs and benefits, this test asks whether a product is more dangerous than an ordinary consumer would reasonably anticipate. For present purposes, assume the test applies to the design at issue.[147]

As under the risk–utility framework, a subgroup blind application of the consumer expectations test evaluates only aggregate outcomes. Courts assess the incidence of injury across the entire user population and compare the aggregate rate to a maximum acceptable level, wl_{max}. Designs producing aggregate injury rates above that level are deemed defective.

Where injury rates exceed consumer expectations for all users, the result mirrors that under risk–utility analysis: the product is defective for everyone. The analytically revealing case, however, is Scenario B, where the aggregate incidence of whiplash falls below maximum acceptable whiplash rate even though the incidence for female users exceeds that threshold. In this setting, aggregate analysis again masks subgroup harm. Because lower injury rates among male users reduce the overall average, the design is treated as consistent with consumer expectations, notwithstanding that it exposes women to levels of risk that exceed what consumers would tolerate for themselves.

Unlike the risk–utility test, this result does not depend on judgments about cost, feasibility, or engineering tradeoffs. Instead, it reflects the assumption that unequal safety outcomes across users are consistent with ordinary consumer expectations so long as average performance appears acceptable. A subgroup blind consumer expectations analysis thus obscures subgroup harm by normalizing materially unequal risk as compatible with reasonable consumer understanding.

Figure 2.



[147] Some jurisdictions hold that the consumer expectations test does not apply to complex products. *See infra* note 120 and accompany test.

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Taken together, the risk–utility and consumer expectations tests reveal a shared doctrinal vulnerability. Although they operate through different mechanisms—balancing costs and benefits on the one hand, and measuring expectations on the other—both can obscure subgroup harms when applied at the aggregate level. Under risk–utility analysis, higher benefits to men can offset serious risks to women; under the consumer expectations test, lower injury rates among men can render elevated risks to women legally invisible. In both frameworks, the failure to consider subgroup outcomes allows products to be deemed non-defective even when they are demonstrably unsafe for a substantial class of ordinary users. This convergence highlights a deeper doctrinal problem: when design defect analysis systematically ignores subgroup considerations, it risks legitimizing products that embed unequal risk into their very design.

B. Towards a Robust Subgroup Considerations Framework

This Section develops the central claim of the Article: that design defect doctrine can and should incorporate subgroup considerations. It proceeds in two steps. First, it argues that courts should explicitly account for subgroup outcomes in design defect analysis—what this Article terms limited subgroup considerations. Incorporating such considerations would prevent designs that generate substantial benefits for men from obscuring predictable harms to women. Second, and more ambitiously, this Part argues for the incorporation of robust subgroup considerations into design defect doctrine. Under this approach, courts would treat differential harm across identifiable subgroups as independently relevant to defectiveness. Rather than asking only whether a product is defective for a particular subgroup, courts would also consider whether a product's design systematically disadvantages women as part of the overall defect analysis. Recognizing such disparities does not require courts to equalize outcomes across all users, but only to account for foreseeable and persistent risk differentials that result from baseline design choices in products marketed as gender neutral.

Before turning to the heart of this analysis, it is important to emphasize that existing design defect law is not, in principle, limited to aggregate analysis. Nothing in products liability law forbids courts from considering whether a product marketed to a broad population is defective for a significant and foreseeable subgroup of users.[148] Neither the

[148] Notably this inquiry is different that what was explored in Part II.B.3, which explained that courts are willing to find a subgroup the ordinary user of a product when the product is specifically marketed to that subgroup. The ordinary user of a child's toy is a child. This analysis still included aggregate effects but for not the entire population only for the marketed subgroup.

risk-utility test nor the consumer expectation test requires defectiveness to be judged solely by aggregate effects across all users. Both tests, however, require the product to be defective for the ordinary user, not an idiosyncratic one. An idiosyncratic user is typically understood as someone who, due to an unusual susceptibility, is injured by a product that is otherwise safe for most consumers—classic examples include individuals with rare allergies or hypersensitivities.[149]

Although the boundary between ordinary and idiosyncratic users has never been sharply defined, early cases generally required plaintiffs to show that an appreciable number of consumers faced a foreseeable risk.[150] Where only a small number of hypersensitive individuals were affected, courts often denied recovery on the ground that the product remained fit for its ordinary purpose.[151] Over time, however, courts became more willing to require warnings for even relatively small subsets of vulnerable users when manufacturers knew or should have known of the risk.[152]

The subgroup of women at issue in this Article does not fall within the category of idiosyncratic users.[153] Women constitute more than half the adult population and represent a clearly identifiable class of ordinary users.[154] Yet despite the doctrinal space for considering subgroup harms, courts rarely do so. It remains the exception, not the rule, for courts to evaluate defectiveness with reference to subgroup outcomes rather than aggregate effects.[155]

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149 *Green v. Smith & Nephew AHP, Inc.*, 629 N.W.2d 727, 753 (Wisc. 2000).

150 *Grau v. Procter & Gamble Company*, 324 F.2d 309 (5th Cir. 1963) (applying Alabama law and noting a substantial number of persons); *Magee v. Wyeth Laboratories*, 214 Cal. App.2d 340 (1963) (substantial number of persons); *Casagrande v. F. W. Woolworth Company,* 340 Mass. 552 (1960) (appreciable number of persons); *Crotty v. Spartanburg's-New Haven, Inc.*, 147 Conn. 460 (1960) (appreciable number of people).

151 *See, e.g.*, *Maynard v. Snapchat, Inc.*, 313 Ga. 533, 536 (2022) (product designer's "duty extends only to reasonably foreseeable risks of harm");

152 *Braun v. Roux Distrib. Co.*, 312 S.W.2d 758, 764-65 (Mo.1958);

153 Yet automakers—and even NHTSA—have long resisted incorporating a 50th-percentile female dummy in crash tests, reasoning that men more likely to suffer injuries because men drive, on average, drive more miles per year and historically engage in riskier behaviors such as drunk driving or speeding. That justification, however, is deeply flawed. There is evidence that these trends are changing and recent studies demonstrate the gap in crash involvement for men and women is narrowing. The number of crashes per annual vehicle miles traveled is higher for women than men—1.52 versus 1.26 injurious crashes per million vehicle miles traveled for female and male drivers respectively. DR Mayhew et al., *Trends in Fatal Crashes Involving Female Drivers*, 35 ACCID ANAL PREV. 407 (2003). As studies demonstrate, decreasing differences in crash involvement for the two sexes indicate increasing exposure and socioeconomic changes in driving behavior among female drivers. Viginia W. Tsai, Craig L. Anderson & Federico E. Vaca, *Young Female Drivers in Fatal Crashes: Recent Trends, 1995–2004*, 9 TRAFFIC INJ. PREV. 65 (2008). Therefore, despite a higher fatality involvement rate for male drivers, traffic trends indicate that future female drivers may be equally as likely to be involved in a crash. Eduardo Romano, Tara Kelley-Baker & Robert B. Voas, *Female Involvement in Fatal Crashes: Increasingly Riskier or Increasingly Exposed?*, 40 ACCID. ANAL. PREV. 1781, 1781 (2008). Regardless, the fact that men have been historically riskier drivers, does not justify designing vehicles that disproportionately endanger women.

154 CONGRESSIONAL RESEARCH SERVICE, STATISTICS ON WOMEN IN NATIONAL GOVERNMENTS 1, DEC. 10, 2024.

155 James A. Henderson, Jr. and Aaron D. Twerski, *Drug Design Liability: Farewell to Comment k*, 67 BAYLOR L. REV. 521, 533 (2015) (noting that "courts approach the relevant benefit/risk balancing in aggregative fashion" in design defect cases and that "courts routinely reject[] arguments on behalf of a small minority of users in order to advance the interests of the large majority.").

This Section therefore turns to a set of cases in which courts have incorporated limited subgroup considerations by focusing the defect analysis on a subgroup of ordinary users. Courts are most likely to consider subgroup harm in three contexts: prescription drugs and medical devices, allergies, and children.[156]

The clearest example of explicit subgroup-based analysis appears in the treatment of prescription drugs and medical devices under *Restatement (Third) of Torts: Products Liability* § 6(c). Jurisdictions adopting § 6(c) deem a prescription product defectively designed only if no reasonable health care provider, aware of the foreseeable risks and benefits, would prescribe it to any class of patients.[157] If the risk–benefit profile is favorable for some subgroup of patients, the product is not defective—even if it is unfavorable for all others.[158] Section 6(c) thus explicitly requires subgroup-based analysis, albeit in a direction opposite to that proposed here: it permits products to escape liability when subgroup benefits exist, rather than imposing liability when subgroup harms persist.

Courts have also, on occasion, treated allergic reactions as evidence of a design defect. In *Green v. Smith & Nephew AHP, Inc.*, for example, the Wisconsin Supreme Court upheld a jury verdict finding latex gloves defectively designed where between five and twelve percent of users experienced allergic reactions ranging from skin irritation to anaphylaxis, and where the gloves contained latex protein levels far exceeding those of competing products.[159] Similarly, courts have been more willing to assess defectiveness with reference to child-specific risks where children will foreseeably interact with the product marketed primarily to adults in ordinary user environments. In *McCormack v. Hankscraft Co.*, a court found a hot-water vaporizer defectively designed because it foreseeably allowed scalding water to gush from a wide opening when tipped over in a child's room.[160]

In the latter two contexts—children and allergies—courts more frequently conceptualize subgroup harm through failure-to-warn theories than design defect. As Judge Sykes observed in her dissent in *Green*, allergy cases "fit most readily into the failure-to-warn

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156 In the second two set of cases, children and allergies, it is possible to conceptualize courts as engaging in an implicit aggregate calculus—summing risks and benefits across the user population and concluding that harms to allergic individuals or children outweigh benefits to others. Courts, however, do not invoke this language, and in some cases it is difficult to believe that a true aggregate analysis would render the product defective. More conservatively, allergy and child cases can be understood as contexts in which courts are more willing to treat subgroup risks as offsetting general utility. More aggressively, they represent domains in which courts are simply more open to considering subgroup harms at all.

157 Unlike § 402A of the Second Restatement which nearly all states embraced by the 1970s, the Third Restatement's product liability § 6 adoption has been limited. Dustin R. Marlowe, *A Dose of Reality for Section 6(c) of the Restatement (Third) of Torts: Products Liability*, 39 GA. L. REV. 1445, 1446, 1449 (2005).

158 Health care providers should only prescribe the prescription drug or medical device to those subsets of patients that have the favorable risk-benefit profile *Restatement (Third) of Torts: Prods. Liab*. § 6 cmt. f.

159 629 N.W. 2d 727, 788 (Wis. 2001). By contrast, courts routinely dismiss claims based on rare allergic reactions. In *Mountain v. Procter & Gamble Co.*, three verifiable allergic reactions out of 225 million bottles of Head & Shoulders shampoo were held insufficient to support strict liability, 312 F.Supp. 534, 537 (E.D. Wis 1970).

160 154 N.W.2d 488 (Minn. 1967).

category" because a product dangerous only to those with allergic sensitivities "cannot be considered dangerous when put to ordinary use by an ordinary consumer without such sensitivity."[161] A similar logic applies to children interacting with adult-marketed products, especially when the harm arises from child's foreseeable misuse of the product rather than a child foreseeably interacting with a product in its ordinary environment—as in *McCormack*, the hot-water vapor case. Consistent with this approach, a number of courts have held that manufacturers of disposable lighters need only provide warnings and are not required to make such products child-safe, even when child misuse is foreseeable.[162] Given children's heightened vulnerability, it is therefore unsurprising that Congress has frequently intervened to mandate design-based protections for children, displacing aggregate or average-user safety analysis and supplementing warning-based regimes.[163]

Tort's law preference to utilize failure-to-warn theories to address subgroup harms, however, map poorly onto gender-based design claims. First, women are not a small or marginal subgroup. They comprise roughly half of the population and often a majority of the relevant market. For example, women purchase more than sixty percent of new automobiles and influence a far larger share of purchasing decisions.[164] Warning-based approaches are ill-suited where warnings would be necessary for a substantial portion and in the case of women usually over half of users. Second, unlike children interacting incidentally with adult products, women are the intended consumers of the products at issue here. Manufacturers market these products as gender neutral while designing them around male bodies. That choice supplies an independent reason to account for subgroup harms directly within the design defect analysis rather than relegating them to warnings.

Outside of these three contexts, courts have occasionally considered subgroup harms but without consistency or doctrinal clarity.[165] Taken together, these cases demonstrate that

161 *Green v. Smith & Nephew AHP, Inc.*, 629 N.W.2d 727, 766 (Sykes, J., dissenting) (Wis. 2001).

162 *Todd v. Societe BIC, S.A.*, 21 F.3d 1402 (7th Cir. 1994); *Kirk v. Hanes Corp.*, 16 F.3d 705 (6th Cir. 1994); *Adams v. Perry Furniture Co.*, 497 N.W. 2d 514 (Mich. Ct. App. 1993). *See* M. Stuart Madden, *Products Liability, Products for Use by Adults, and Injured Children: Back to the Future*, 61 TENN. L. REV. 1205, 1220–1222 (1994) (discussing disposable lighter cases).

163 *See, e.g.*, 15 U.S.C. §§ 1471–1477 (mandating child-resistant packing for prescription drugs, household chemicals, and pesticides); 15 U.S.C. §§ 1261–1278 (requiring special treatment for products likely to be accessible to children).

164 Women Andrew Boyle, Men vs. Women: The Gender Divide of Car Buying, CJ Pony Parts (2024) *available at* https://www.cjponyparts.com/resources/men-vs-women-car-buying.

165 For example, courts have intermittently considered gender-specific risks when the alleged defect manifests differently across sexes. In *Ingham v. Johnson & Johnson*, twenty-two women alleged that Johnson & Johnson's talcum powder caused ovarian cancer—a harm unique to women, 608 S.W.3d 663 (Mo. Ct. App. 2020) (upholding 2 billion of the 4.7 billion a Missouri jury initially awarded the women). Internal company documents revealed that Johnson & Johnson had long known about asbestos contamination risks, but suppressed this information. *Id.* Although the product was marketed broadly, the court effectively treated women as the relevant class of ordinary users. Even less frequently, courts have found products defective for subgroups defined by size or weight. In *Uxa ex rel. Uxa v. Marconi*, a child's car seat marketed for children weighing between 22 to 70 pounds was found defective when a toddler at the lower end of that range suffered severe head injuries, 128 S.W.3d 121, 126–127 (Mo. Ct. App. 2003). Expert testimony established that smaller children required more head protection than the seat provided, *id.* at 127.

limited subgroup analysis is not foreign to products liability doctrine, even if courts have been hesitant to apply it and rarely permit such claims to proceed. Conceptually, they confirm that a product need not be unsafe for all users to be defective. Moreover, there are independent reasons to account for explicit subgroup harms in the gender-design cases.

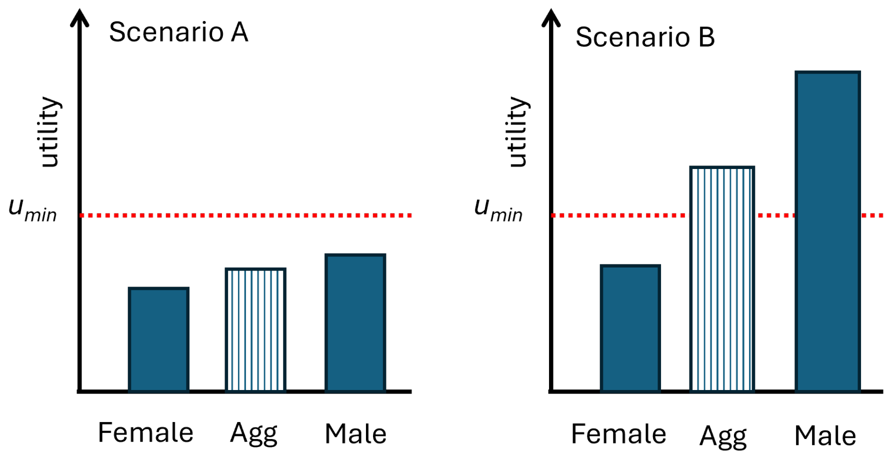
At a minimum, courts should more consistently and explicitly embrace this limited form of subgroup analysis by recognizing subgroup harms as relevant to defectiveness where the subgroup constitutes a substantial and foreseeable class of ordinary users. What distinguishes male-default design is therefore not doctrinal novelty, but doctrinal omission: courts have failed to extend this existing logic to sex-based design baselines for products marketed to the general public, despite women constituting a majority of adult consumers and a plainly foreseeable class of users.

By contrast, the incorporation of robust subgroup considerations—the explicit evaluation of differential safety profiles for men and women as an independent factor in design defect analysis—would represent a more substantial doctrinal shift. Under this approach, courts would move beyond merely recognizing subgroup harms to treating systematic disparities in safety outcomes as independently relevant to defectiveness, marking a genuinely new development in products liability law. Importantly, this approach is confined to products marketed as gender neutral, where foreseeable and persistent safety disparities affecting a substantial class of ordinary users arise from baseline design choices rather than isolated misuse or atypical physiology.

1. Risk-Utility Test

To more crisply illustrate the conceptual difference between a products liability claim that excludes all subgroup considerations and one that allows for a product to be defective for a subgroup whose risk-utility profile differs from the aggregate population, reconsider the car seat example from Section A. Again, assume that the benefits and risks associated with the design of a product can be represented by one measure of utility, u. Figure 3 shows the utility of a car seat design for female, male, and average occupants. The red dashed line marks the minimum utility threshold; designs falling below it are defective.

Figure 3.



The analytically significant case arises in Scenario B, where utility differs systematically across subgroups. In this scenario, the aggregate utility of the car seat exceeds minimum utility threshold even though the utility for female users falls below that threshold. Because higher utility for male users offsets the elevated risk to women, a subgroup blind application of the risk–utility test treats the design as non-defective. Aggregate analysis thus obscures the fact that the product imposes unacceptable risk on a substantial class of ordinary users.

When courts incorporate limited subgroup considerations, this result changes. Rather than relying exclusively on aggregate utility, courts may evaluate whether the product's utility falls below acceptable levels for a particular subgroup. Under this approach, women could establish that the car seat is defectively designed as to them, even though the design appears acceptable when evaluated only at the population level.

More ambitiously, the risk–utility test may be extended to incorporate robust subgroup considerations. Under this approach, the inquiry moves beyond whether utility for each subgroup satisfies minimum thresholds and instead examines the disparity in utility across subgroups. Where a product marketed to a broad population provides materially greater safety benefits to men than to women, that disparity itself becomes relevant to the assessment of design reasonableness.

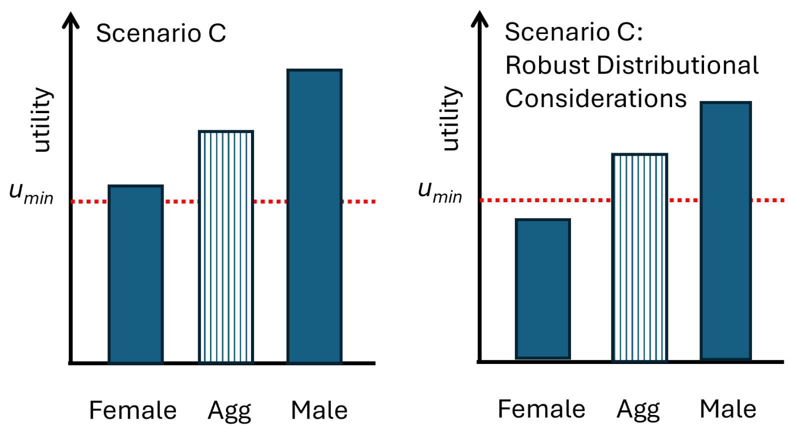
Importantly, accounting for such disparities does not transform products liability into a general equality regime. Robust subgroup analysis is confined to products marketed as gender neutral, where foreseeable and persistent safety disparities affecting a substantial class of ordinary users arise from baseline design choices rather than isolated misuse or atypical physiology. In these circumstances, differential allocation of safety benefits may properly inform whether a design is unreasonably dangerous under the risk–utility framework.

To operationalize a robust subgroup analysis under the risk-utility test, factfinders would weigh evidence of differential harm (or the benefit of products with similar risk-utility profiles) alongside traditional factors such as cost, feasibility, risks and benefits of the design. Whether a design is defective would continue to depend on the overall balance of factors, with subgroup disparities weighed in relation to other considerations in the analysis.

Figure 4 illustrates how this approach operates. As in the prior analysis, utility is represented by u, and the design is defective if utility falls below a minimum threshold, u_{min}. In Scenario C, the utility of the standard car seat exceeds the threshold for both female and male users when subgroup outcomes are evaluated separately. Under a framework limited to subgroup-specific thresholds, the design would not be defective. Yet the car seat provides materially greater utility to male users than to female users, reflecting a substantial disparity in protection.

When robust subgroup considerations are taken into account, this disparity becomes relevant to the defect analysis. Rather than asking only whether the product clears minimum safety thresholds for each subgroup, the factfinder may consider whether the difference in utility across users is sufficiently pronounced to render the design unreasonable. In Scenario C, a court could assign negative weight to the disparity in utility between female and male users, such that the effective utility for women falls below the minimum threshold, rendering the design defective as to them. This result does not follow automatically from disparity alone; it depends on how subgroup considerations are weighed relative to other factors in the risk–utility calculus.

Figure 4.



Determining the appropriate weight to assign subgroup considerations is inherently context dependent and difficult. This Article does not purport to identify a fixed or optimal weighting rule, leaving that question for future scholarship and doctrinal development. Nevertheless, there are several considerations that a jury may contemplate when evaluating the significance of subgroup considerations in a products liability case, including the manufacturer's design choices and the composition of the product's user population.

One potentially relevant consideration is the extent to which the manufacturer incorporated women into the design baseline. Intentionality in this context is not binary. Where a manufacturer designs a product around male physiology alone while marketing it to both men and women, a jury may reasonably treat even relatively modest subgroup disparities as probative of defectiveness. The significance of such disparities reflects not a judgment about motive, but an assessment of whether the design choices reasonably accounted for foreseeable users. By contrast, where a manufacturer partially, but incompletely, accounts for women's physiology in the design process while marketing the product to both sexes, a jury may reasonably require a more substantial subgroup disparity before drawing an inference of defectiveness.

At one end of the spectrum are products whose designs account for women but do so in an incomplete or underinclusive manner. Prescription drugs illustrate this dynamic.[166]

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[166] *Restatement (Second) of Torts* § 402A cmt. k addressed the products liability claims and prescription drugs. Commentators have noted that "the confused language of Comment k has spawned chaos in the decision law of drug design liability." James A. Henderson, Jr. and Aaron D. Twerski, *Drug Design Liability: Farewell to Comment k*, 67 BAYLOR L. REV. 521, 543 (2015). Professors Henderson and Twerski note relying on comment k courts have taken at least eight different approaches to drug design liability including holding "that manufacturers of prescription drugs are entitled to escape liability for drug designs completely; . . .that it is the defendant's burden to prove that a drug's benefits outweigh its risks; . . . and that plaintiffs can establish liability if an alternative FDA-approved drug is as effective as, and safer than, the drug in question."). *Restatement (Third) of Torts: Prods. Liab.* § 6(c) preserves the ability to design defect claims for prescription drugs but heightened the requirements to do so. More specifically, § 6(c) finds a prescription drug defective only if a health care provider

Although women's participation in clinical trials has increased substantially since the mid-1990s, meaningful gaps remain, particularly in certain therapeutic areas.[167] Continued underrepresentation has resulted in dosing guidelines and safety profiles that remain calibrated largely to male physiology.[168] As a result, a jury may assign less weight to subgroup disparities in evaluating the significance of resulting subgroup harms of prescription drugs in comparison to other products that are designed solely around male bodies.[169]

At the opposite end of the spectrum are products whose design processes affirmatively center male bodies. Automobile design lies close to this pole. Vehicle safety systems have long been developed and validated using crash-test dummies modeled on the average male body. Manufacturers that limit their testing to federally mandated scenarios necessarily design around a male baseline, because those mandates historically center male anthropometry. Even the use of the 5th-percentile female dummy—essentially a scaled-down male model—continues to privilege male physiology. In this context, adherence to regulatory minima may reasonably be understood as an intentional choice to calibrate safety systems to male bodies, notwithstanding the absence of discriminatory motive.

By contrast, manufacturers that supplement regulatory testing with female-specific models—such as 50th-percentile female dummies or computational simulations incorporating female biomechanics—demonstrate an effort to account for sex-based differences in injury mechanisms.[170] This distinction allows courts to differentiate between passive reliance on male-centered standards and affirmative design choices that address foreseeable risks to female users.

A second relevant consideration is the proportion of women among the product's users. Courts could permit juries to weigh subgroup harms in relation to women's share of the market. On this view, the larger the percentage of female users, the greater the relevance of subgroup disparities to the defect analysis. A product utilized by predominantly to women but designed around male bodies presents a more compelling case for liability than a product used only incidentally by women, even if the magnitude of risk disparity is comparable. This

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knowing the foreseeable risks and benefits of the drug would not prescribe the drug to any class of patients. *Restatement (Third) of Torts: Prods. Liab.* § 6(c).

167 Karen M. Goldstein, et al., *Strategies for Enhancing the Representation of Women in Clinical Trials: An Evidence Map*, 13 SYSTEMATIC REV. 2, 2 (2024).

168 *See supra* note 88 and accompanying text.

169 That attenuation is not absolute. Drug manufacturers can further mitigate concerns about intentionality by taking proactive steps to account for sex-based differences, by adopting enrollment practices that facilitate women's participation, such as offering weekend, evening, or virtual study visits. These measures demonstrate an effort to address foreseeable disparities and reduce the extent to which male physiology continues to structure drug design.

170 See the discussion of Volvo below, which is an industry leader in incorporating women into the design of cars. *See infra* notes **Error! Bookmark not defined.**–192and accompanying text.

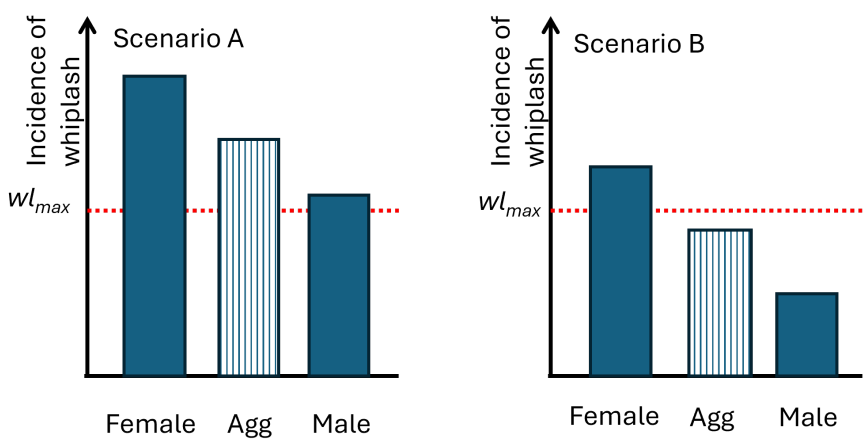
approach aligns defect analysis with market realities and reinforces the principle that products should be reasonably safe for the users to whom they are sold.

Together, these considerations confirm that subgroup analysis can be integrated into design defect doctrine without rigid rules or bright-line thresholds. Instead, they provide juries with structured, familiar tools for evaluating the reasonableness of design choices in light of foreseeable risks to identifiable users—an inquiry that lies at the core of modern products liability law.

2. Consumer Expectation Test

To illustrate how subgroup considerations operate under the consumer expectations test, consider again the car-seat design discussed in Section A. Under this framework, a product is defective if it is more dangerous than an ordinary consumer would reasonably expect when used in a foreseeable manner. In this context, the relevant inquiry is whether the incidence of whiplash exceeds a maximum acceptable level, wl_{max}.

Figure 5.



The analytically significant case arises in Scenario B, where whiplash incidence differs systematically across subgroups. In this scenario, the aggregate incidence of whiplash falls below maximum incidence of whiplash, even though the incidence for female users exceeds that threshold. Because lower injury rates among male users reduce the overall average, a subgroup blind application of the consumer expectations test treats the design as non-defective. Aggregate analysis thus obscures the fact that the product exposes women to levels of risk that exceed what consumers would reasonably tolerate for themselves.

When courts incorporate limited subgroup analysis, this result changes. Rather than collapsing subgroup outcomes into a single average, courts may consider whether the product fails to meet reasonable safety expectations for a substantial class of ordinary users. Under this approach, women could establish that the car seat is defective as to them, even though the design appears acceptable when evaluated only at the aggregate level.

The consumer expectations inquiry could also be adapted to account for robust subgroup considerations. Under this approach, the analysis would extend beyond whether an ordinary consumer would expect injury rates to remain below a specified threshold to include whether consumers would anticipate materially different risk profiles across identifiable groups of users. In this context, courts could consider whether a product marketed as gender neutral imposes substantially disproportionate risks on women relative to men. Framed in this way, the consumer expectations test would not depart from its focus on reasonable consumer understanding but would incorporate attention to whether consumers reasonably expect products to allocate safety risks unevenly across ordinary users. Under this reframing, the car seat in Scenarios A and B could be deemed defective not only because whiplash incidence exceeds acceptable thresholds for women, but also because reasonable consumers would not expect safety features to impose materially disparate risks across users.

C. Reasonable Alternative Design

This Section builds on the framework developed in Section B by examining how subgroup analysis interacts with the reasonable alternative design requirement under the risk–utility test. Consider again the standard car seat discussed in Section A, which was designed using a male crash-test dummy and results in significantly higher whiplash rates for women than for men. Assume that three alternative car seat designs are available, each of which substantially reduces or eliminates the disparity in whiplash outcomes between female and male occupants at nominal additional cost.

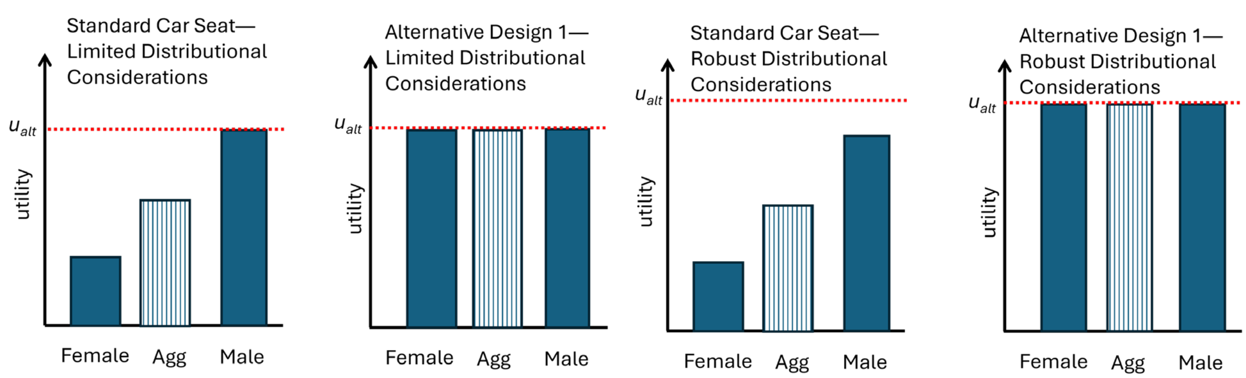
Alternative Design 1 eliminates the disparity by reducing women's whiplash rates to match those experienced by men under the standard design. Alternative Design 2 eliminates the disparity by increasing men's whiplash rates to match those experienced by women under the standard design. Alternative Design 3 eliminates the disparity by reducing whiplash rates for women while modestly increasing whiplash rates for men relative to the standard design.

Although each alternative design equalizes outcomes across sexes—a factor that weighs positively under a subgroup sensitive framework—that fact alone does not determine defectiveness. Whether the standard design is defective depends on the comparative utility of the alternative designs, evaluated in light of cost, feasibility, safety, and subgroup considerations.

1. Alternative Design 1

Figure 6 illustrates the utility of the standard car seat relative to Alternative Design 1 for women, men, and the aggregate user. As in the prior analysis, assume the benefits and risks associated with the design of a product can be represented by one measure of utility, u. The standard car seat is defective if its utility falls below that of the alternative design, u_{alt}. When robust subgroup considerations are taken into account, as outlined in Section B, the design defect analysis permits consideration of utility for female and male users separately rather than focusing exclusively on the average user and also positively weights the alternative design equality of utility profiles for female and male users.

Figure 6.



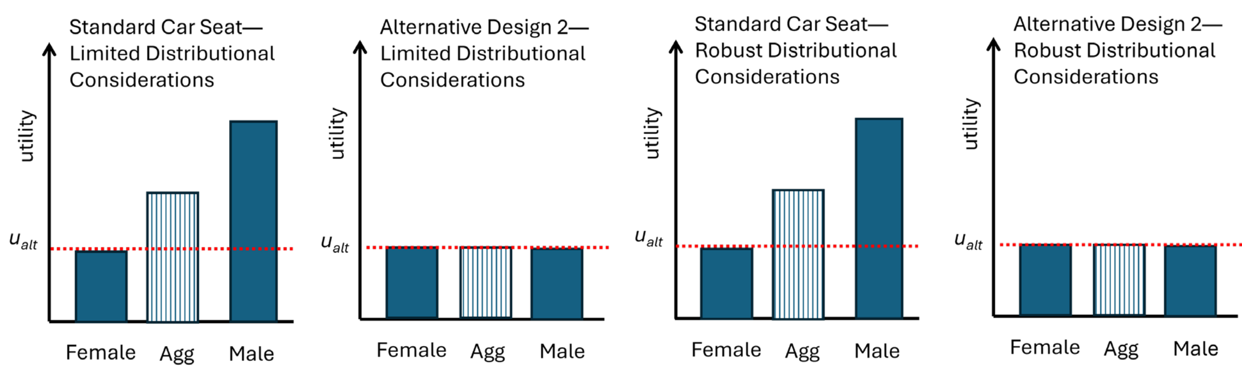
Alternative Design 1 represents the easy case. This design eliminates the disparity in whiplash rates between men and women by decreasing women's incidence of whiplash to match those of men—that is, by increasing the utility of the car seat for female occupants to match that of male occupants. When limited subgroup considerations are applied, the standard car seat is defective as to female occupants and the aggregate user, but not as to male occupants. Because the new framework gives additional positive weight to products whose safety profiles are roughly equivalent for men and women, the utility for the alternative design shift upwards once robust subgroup considerations are taken into account. Under the new framework, the alternative design yields higher utility for both women and men, the standard car seat is defective for all users.

Importantly, incorporating women into the design process will likely result in the design of products that will increase the utility for all end users. For example, the incorporation of the 5th-percentile female dummy in crash tests help to speed the adoption of curtain airbags—airbags that deploy from the roof rail—and torso airbags—which deploy from the side of a door—during collisions. Curtain and torso airbags have been estimated to decrease fatalities in near-side impact collisions by over 30% for all users, not just women.[171]

2. *Alternative Design 2*

Figure 7 illustrates the utility of the standard car seat relative to Alternative Design 2, which eliminates the disparity in whiplash rates between female and male users by increasing men's whiplash rates to match those experienced by women under the standard car seat—that is, by lowering the utility of the car seat for male occupants to match that of female occupants.

Figure 7.



Accounting for robust subgroup considerations does not change this result from only accounting for subgroup harm. Because the alternative design offers no utility improvement for women and imposes a significant utility loss on men, little to no weight should be given to the fact that this alternative design has similar safety profiles for men and women. As a result, there is virtually no difference in the framework that accounts for limited and robust subgroup considerations. Either way, the standard car seat should not be defective.

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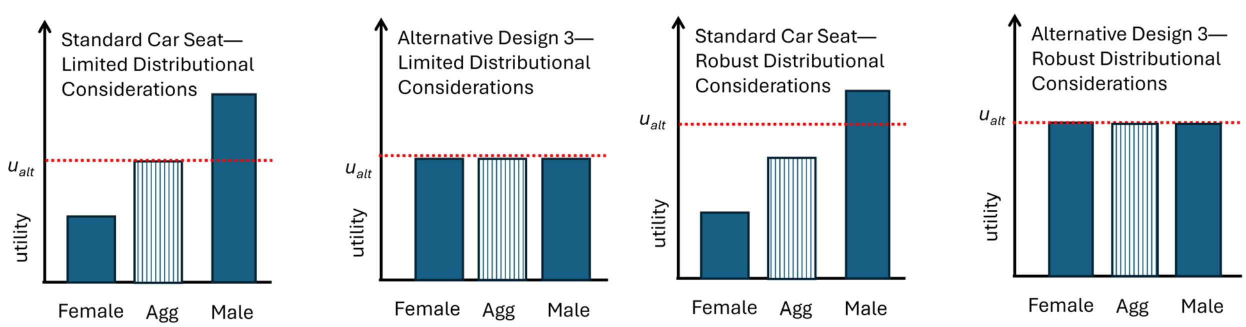
[171] NHTSA, UPDATED ESTIMATES OF FATALITY REDUCTION BY CURTAIN AND SIDE AIR BAGS IN SIDE IMPACTS, AND PRELIMINARY ANALYSES OF ROLLOVER CURTAINS (2014) at v, *available at* https://crashstats.nhtsa.dot.gov/Api/Public/ViewPublication/811882.pdf.

Few products are likely to fall squarely into this category. The inclusion of females in the design process is far more likely to lead to improved safety and functionally for women than products with the utility profile of Alternative Design 2. Moreover, products exhibiting the safety profile of Alternative Design 2 are likely products in which a single, uniform design cannot reasonably accommodate all users.[172] In such cases, offering multiple sizes or configurations—rather than degrading performance for one group—may be the appropriate design response. Surgical instruments, given well-documented differences in hand size and grip strength, provide a common example.[173]

3. Alternative Design 3

Figure 8 compares the standard car seat with Alternative Design 3, which reduces whiplash rates for women while modestly increasing whiplash rates for men—that is, by lowering the utility of the car seat for male occupants and elevating the utility of the car seat for female occupants. Under a framework limited to subgroup-specific thresholds, the standard design is defective for women but not the average user. However, when robust subgroup considerations are incorporated, the analysis changes.

Figure 8.



Because this framework assigns positive weight to designs that provide more comparable safety profiles across users, the utility of Alternative Design 3 increases relative to the limited subgroup analysis. In this robust subgroup scenario, the standard car seat is defective for women and the aggregate user as well. This example highlights that improving

172 Several factors will affect whether a product will need to be offer in multiple sizes such as the degree to which a product must fit the user's body closely for proper function or comfort and the characteristics of the target user group and context of user. The tighter fit requirements usually demand multiple sizes. Products intended for a broad and diverse audience generally require more size variation.

173 Riegel v. Medtronic, Inc., 552 U.S. 312, (2008) held only that Class III medical devices that receive premarket approval from the FDA are preempted from design defect claims. The vast majority of surgical instruments are Class I devices. *Id.* 316–17. For a discussion of preemption doctrine in products liability *see* Catherine Sharkey, *Products Liability Preemption: An Institutional Approach*, 76 GEO. WASH. L. REV. 449 (2008); Catherine M. Sharkey, *Federalism Accountability: "Agency-Forcing" Measures,* 58 DUKE L.J. 2125 (2009).

safety for one group may sometimes require modest tradeoffs for another, and that the permissibility of such tradeoffs depends on how subgroup considerations are balanced against other risk–utility factors.

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Recognizing subgroup harms within design defect analysis would affect manufacturers' ex ante assessment of design reasonableness in predictable ways. When liability turns in part on whether a product allocates safety risks unevenly across ordinary users, manufacturers would have incentives to consider a broader range of foreseeable users at the design stage. That shift follows directly from familiar tort principles. Design defect liability has long encouraged manufacturers to internalize the foreseeable costs of their design choices, and accounting for subgroup harms simply makes visible a category of risk that aggregate analysis currently obscures.

Importantly, this incentive effect does not require courts to mandate particular design outcomes or to impose a general duty of universal design. It requires only that manufacturers account for the foreseeable consequences of baseline design decisions that privilege one group of users over others in products marketed to a broad population. In this respect, a subgroup attentive design defect framework operates through conventional mechanisms of fault, foreseeability, and reasonableness, rather than through regulatory command. Any resulting changes in design practice are thus a byproduct of adjudication, not a departure from established products liability doctrine.

IV. ADDITIONAL CONSIDERATIONS

This Part addresses several doctrinal issues that arise when applying products liability law to claims involving male-default design, focusing in particular on causation and the reasonable alternative design requirement. Finally, this Part concludes by addressing the scope of the Article's focus on women.

A. CAUSATION

Causation can present analytical questions in products-liability suits alleging that male-default design paradigms disproportionately endanger women. Traditional defect claims often involve a discrete design choice or manufacturing flaw that produces an identifiable failure. Gendered design harms, by contrast, more commonly arise from a structural omission: the systematic exclusion of female anthropometric and biomechanical data during the design process. In such cases, the alleged injury does not result from a malfunctioning component, but from a product whose baseline calibration reflects male physiology as the reference point. As a matter of cause in fact, plaintiffs must still show that their injuries would not have occurred but for that design choice.

In practice, plaintiffs can meet this burden using familiar evidentiary tools. In automobile cases, crash reconstruction, biomechanical testimony, and comparative design evidence are routinely used to link particular injuries to particular design calibrations.[174] Expert testimony can explain how restraint systems or airbags optimized for male proportions interact differently with bodies of different stature, mass distribution, and tissue geometry, thereby connecting population-level findings to an individual plaintiff's injury.

Much of the evidence supporting such claims is necessarily population based. A substantial body of research documents higher rates of whiplash, thoracic injury, and fatality for women in otherwise comparable automobile crashes, suggesting that male-default design systematically elevates risk for female occupants.[175] Although statistical disparities alone may be insufficient to establish traditional but-for causation in every case,[176] tort law has long recognized that strict outcome-determinative proof may be ill suited where a defendant's conduct alters the probability of injury rather than directly producing harm.

Courts have addressed this problem in other contexts through limited forms of probabilistic causation. Lost chance doctrine in medical malpractice treats the loss of a statistically measurable opportunity—such as survival or recovery—as a cognizable injury and apportions liability in proportion to the probability lost.[177] Increased-risk litigation,[178] such as medical monitoring claims[179] or fear-of-future- disease claims,[180] similarly recognize liability for scientifically demonstrable risk creation even when traditional more-likely-than-

174 *See, e.g.*, Eskin v. Carden, 842 A.2d 1222, 1225–1226 (Del. 2024) (holding that the trial court may admit "qualified biomechanical expert testimony regarding the physical forces involved in automobile accidents and the effect on the human body those forces may produce . . ."); Dean v. Cherokee Ins. Co., Royal Trucking Co., 2021 U.S. Dist. LEXIS 260660, (N.D. Ga. 2021) (admitting testimony of an expert in an "area of specialty in engineering mechanics in crash reconstruction and the effects of force and acceleration on the human body."); Dawson v. Chrysler Corp., 630 F.2d 950, 953–54 (3d Cir. 1980) (discussing the trade-off between the design adopted the alternative design proposed by the plaintiff).

175 *See infra* Part I.A.2.

176 *In re Joint E. & S. Dist. Asbestos Litig.,* 52 F.3d 1124, 1128–30 (2d Cir. 1995).

177 *Herskovits v. Group Health Coop. of Puget Sound*, 664 P.2d 474 (Wash. 1983) (recognizing loss of chance as a compensable injury in medical malpractice). See generally, Lauren Guest, David Schap & Thi Tran, *The "Loss of Chance" Rule as a Special Category of Damages in Medical Malpractice: A State-by-State Analysis*, 21 J. LEGAL ECON. 53 (2015).

178 Gary E. Marchant & Michael S. Baram, *The Use of Risk Assessment Evidence to Prove Increased Risk and Alternative Causation in Toxic Tort Litigation*, 41 FED'N INS. & CORP. COUNS. Q. 95 (1990).

179 *See* Robert P. Charrow, Medical Monitoring, as Proposed in the Restatement (Third) of Torts, Raises Scientific and Other Issues that Warrant Attention, 64 Jurimetrics 3 (2025) (discussing the tort and that there is some disagreement as to whether 17 or 14 states have adopted the tort).

180 *See, e.g.*, Keith W. Lapeze, *Recovery for Increased Risk of Disease in Louisiana*, 58 LA. L. REV. 249, 253 (1997).

not causation cannot be established for any individual plaintiff. These doctrines share a common premise: when wrongful conduct systematically worsens a plaintiff's probabilistic prospects, causation should be evaluated in terms of risk contribution rather than outcome determinism.[181]

Claims involving gendered design raise related considerations. Restraint systems calibrated around prevailing anthropometric assumptions do not injure every woman in every crash; rather, they may shift injury risk in systematic and measurable ways. A design that increases the risk of thoracic injury for women may not be provably responsible for any particular plaintiff's injury to a greater-than-fifty-percent certainty. Yet its contribution to risk may be scientifically grounded and traceable to identifiable design choices. As in other probabilistic-causation contexts, the legally relevant harm may lie in the loss of a safer baseline that would have existed under a more inclusive design framework.

Critics may contend that extending probabilistic causation doctrines to design-based disparity claims risks speculative liability, dilutes causation standards, or places undue weight on population-level statistics at the expense of individualized adjudication. On this view, doctrines such as lost chance and increased-risk liability are best understood as context-specific responses to medical malpractice or toxic exposure and should not be generalized to product-design litigation.

These concerns underscore the importance of careful doctrinal boundaries. Properly understood, however, probabilistic causation doctrines do not dispense with individualized proof; rather, they adapt it to reflect evidentiary limits inherent in certain forms of harm. Lost chance, increased-risk liability, and medical monitoring require scientifically demonstrable, defendant-specific contributions to diminished prospects. Design-based disparity claims can fit within this framework where epidemiological evidence is directly linked to concrete, identifiable design choices. Moreover, proportional liability mechanisms can constrain recovery to the extent of the risk attributable to the defendant's conduct, preserving a meaningful connection between conduct and damages.

Recognizing probabilistic causation in this setting would not represent a wholesale departure from established tort principles. Instead, it reflects a continuing doctrinal effort to address situations in which measurable risk increases cannot be reduced to deterministic proof for any single plaintiff. In such cases, the law must determine whether evidentiary

[181] Other examples include fear of future disease claims, Keith W. Lapeze, *Recovery for Increased Risk of Disease in Louisiana*, 58 LA. L. REV. 249, 253 (1997), or increased risk litigation, Gary E. Marchant & Michael S. Baram, *The Use of Risk Assessment Evidence to Prove Increased Risk and Alternative Causation in Toxic Tort Litigation*, 41 FED'N INS. & CORP. COUNS. Q. 95 (1990).

uncertainty should preclude recovery altogether or whether carefully cabined probabilistic approaches provide a more accurate account of the harm alleged.

Finally, where a plaintiff establishes that a manufacturer intentionally designed a product around a male baseline while marketing it to both sexes, and that this design choice measurably increased injury risk for women, courts could consider shifting the burden on causation to the manufacturer. Once the plaintiff demonstrates a scientifically supported increase in risk directly traceable to identifiable design calibrations, it would be reasonable to require the manufacturer to rebut the inference that this increased risk contributed to the injury. Tort law has long recognized burden-shifting doctrines in situations involving evidentiary asymmetry or structural uncertainty. Under alternative liability, courts shift the burden where multiple defendants created risk and the plaintiff cannot isolate which caused the harm.[182] Under res ipsa loquitur, courts permit an inference of causation where the instrumentality was under the defendant's control and the injury would not ordinarily occur absent negligence.[183] A calibrated burden-shifting rule in male-default design cases would operate similarly: where the defendant's design baseline systematically elevated risk for a foreseeable class of users, and the plaintiff falls within that class, evidentiary uncertainty should not automatically defeat recovery. Instead, the manufacturer—who possesses superior access to design data, testing protocols, and calibration assumptions—should bear the burden of demonstrating that its baseline choice did not contribute to the injury.

Properly confined to cases involving deliberate reliance on male reference models and scientifically demonstrable risk disparities, such burden shifting would remain consistent with established tort principles while preventing structural design asymmetries from becoming categorical shields against liability.

B. Reasonable Alternative Design

The reasonable alternative design requirement can also raise analytical questions in design defect claims premised on male-default design. In jurisdictions that incorporate a reasonable alternative design requirement into the risk–utility test, plaintiffs are generally expected to identify a safer, feasible design that would have reduced or avoided the alleged harm without impairing the product's utility or imposing disproportionate cost.[184] Scholars

[182] *Summers v. Tice*, 33 Cal. 3d 80, 88 (1948).
[183] *Byrne v. Boadle*, (1863) 159 Eng. Rep. 299 (Exch.).

[184] Scholars and courts disagree on how many states require plaintiffs to prove a reasonable alternative design as part of the risk utility test. *Compare* Aaron D. Twerski & James A. Henderson, Jr., *Manufacturers' Liability for Defective Product Designs: The Triumph of Risk-Utility*, 74 BROOK. L. REV. 1061, 1080–81 (2008) (observing that approximately twenty states require plaintiffs to prove a reasonable alternative design as part of the risk-utility test) *with* John F. Vargo, *The Emperor's New Clothes: The American Law Institute Adorns a "New Cloth" for Section 402A Products Liability Design Defects—A Survey of the*

have observed that this requirement may impose significant evidentiary demands on plaintiffs, potentially limiting one of the traditional functions of strict products liability—namely, easing informational and resource asymmetries that can complicate negligence-based claims.[185] These concerns may be more pronounced in cases involving gendered design, where the absence of women-centered alternatives is itself part of the alleged defect.

One source of difficulty is that alternative designs adequately accounting for female physiology may not be readily identifiable at the time of litigation, not because such designs are infeasible, but because women have historically been underrepresented in design and testing processes. Gaps in female anthropometric, biomechanical, and ergonomic data can complicate efforts to model counterfactual designs and to quantify their potential safety benefits. This challenge may be particularly salient for technologically complex products, such as automobiles or industrial equipment, where design decisions depend heavily on advanced modeling and empirical validation. In these contexts, the alternative design requirement risks insulating male-default designs from liability precisely because manufacturers failed to collect the data necessary to reveal safer options for women.[186]

At the same time, the reasonable alternative design requirement is neither uniform nor rigidly applied. A substantial number of jurisdictions applying a risk–utility framework do not treat proof of a reasonable alternative design as a formal prerequisite to liability.[187] Even among courts that do, the requirement is often applied with flexibility. Plaintiffs are not typically required to present fully engineered prototypes or production-ready designs.[188] Alternative design evidence may instead take more modest forms, including conceptual modifications,[189] adjustments to existing design parameters, or comparisons to products already available in the market.[190]

In some cases, industry practice may further inform the alternative design inquiry. Certain manufacturers have undertaken early efforts to incorporate women into product design, providing examples of feasible alternatives. In the automotive context, Volvo is frequently cited for integrating female crash-test dummies into safety testing well before

States Reveals a Different Weave, 26 U. MEM. L. REV. 493, 537 (1996) (finding that 8 states require the reasonable alternative design to establish a design defect claim) and Delaney v. Deere & Co., 999 P.2d 930, 946 (2000) ("Our own research also reflects that a majority of jurisdictions in this country do not require a reasonable alternative design in product liability actions.").

185 *See supra* notes 134–135 and accompany text.

186 A similar phenomena has been identified in toxic torts, where manufacturers failure to conduct basic safety testing of chemicals enabled manufacturers to escape liability because plaintiffs struggled to conduct such testing on their own. See Wendy E. Wagner, *Choosing Ignorance in the Manufacture of Toxic Products*, 82 CORNELL L. REV. 773, 791–794 (1997).

187 *See supra* note 184.

188 Unrein v. Timesavers, Inc., 394 F.3d 1008, 1012–13 (8th Cir. 2005) (explaining that a reasonable alternative design need not be manufactured as a "new device or protype" and may be shown through expert testimony describing conceptual design modifications).

189 *Id.*

190 *Id.*

regulatory mandates required it and for developing a pregnant crash-test dummy to study the effects of restraint systems on pregnant occupants.[191] These initiatives contributed to the development of Volvo's Whiplash Protection System, which significantly reduces neck and head injuries and substantially narrows sex-based disparities in whiplash outcomes.[192] Volvo's experience demonstrates that gender-inclusive design is both technologically feasible and capable of improving safety outcomes without sacrificing overall utility.

Not all products pose the same evidentiary difficulties. For many products, addressing gender disparities may require relatively straightforward modifications rather than complex redesign. Surgical instruments, for instance, often need only be offered in multiple sizes to accommodate differences in hand size and grip strength.[193] Personal protective equipment may also require multiple size options, adjustable sizing, or modified weight distribution.[194] In these contexts, the reasonable alternative design inquiry does not demand speculative innovation, but the need for variable sizing options.

Finally, evidence that a manufacturer deliberately designed a product around a male baseline while marketing it to both sexes may justify relaxing, or in some cases dispensing with, the reasonable alternative design requirement. That requirement is not intended to reward manufacturers for adopting a male archetype while promoting products as gender neutral; it is meant to assess whether the challenged design reflects a reasonable allocation of risk. Where a manufacturer's design choices systematically externalize risk onto women, the absence of a fully specified women-centered alternative should not categorically foreclose liability. To hold otherwise, would allow historical data gaps and exclusionary design practices to become self-justifying barriers to recovery.[195] In such cases, the practical force

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[191] Steve Tengler, *Volvo Cars Cared About Safety Long Before the #MeToo Movement*, FORBES (Feb. 16, 2021) https://www.forbes.com/sites/stevetengler/2021/02/16/volvo-cars-cared-about-safety-long-before-the-metoo-movement/ (noting that Volvo started utilizing female crash test dummies in the driver's seat before 1996); Volvo, *Volvo Cars Develops World's First Pregnant Crash Test Dummy* (Sept. 18, 2003), https://www.volvocars.com/us/media/press-releases/758BCE211ADE1E23/.

[192] The Whiplash Protection System served as the inspiration for the alternative design carseat utilized in Section III of this Article. This innovative design—featuring a high, close head restraint and a reinforced seat back—provides superior protection. In a rear-end collision, the seat back initially moves in tandem with the occupant's body before reclining to absorb impact more evenly. Volvo developed the Whiplash Protection System (WHIPS) which this alternative seat design is modeled. Studies found the WHIPS decreased initial neck injuries by 33% and long term neck injuries by more than 50%. The injury-reducing effect was higher for women than men. Lotta Jakobsson, Irene Isaksson-hellman & Magdalena Lindman, *WHIPS (Volvo Cars' Whiplash Protection System)—The Development and Real-World Performance*, 9 TRAFFIC INJURY PREVENTION 600 (2008).

[193] Braman & Robertson, *supra* note 64, at 359 (noting that surgical tools need to either be designed with principles of universal design to make tools "work just as well for larger and small hands" or "tools need to be designed in multiple sizes").

[194] Women in Global Health, Fit for Women? Safe and Decent PPE for Women Health and Care Workers, at 46 (Nov. 2021), available at chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://gendro.org/wp-content/uploads/2023/11/1_WGH-Fit-for-Women-report-2021.pdf.

[195] Similar observations of gaps in data have been made in other fields. For example, Professor Wendy Wagner has argued that the burden for producing basic scientific research on the toxicity of chemicals should shift to the manufacturer because as the law currently stands they have the incentive to not produce such data as the lack of data in essence provides immunity to tort litigation. Wendy E. Wagner, *Choosing Ignorance in the Manufacture of Toxic Products*, 82 CORNELL L. REV. 773, 791–794 (1997).

of the alternative design requirement is substantially diminished: where manufacturers have never attempted to design around female anthropometry or to test products across a representative range of users, it is highly likely that a safer, technologically feasible alternative design exists. In such a scenario, it is reasonable to infer that safer, technologically feasible alternatives were available even if the plaintiff cannot fully specify the contours of such an alternative at the time of litigation.

C. WHY FOCUS ON WOMEN

Women are not the only users who may experience disproportionate harm from products designed around an "average" body. Evidence suggests that older individuals, heavier users, and those at the extremes of the size distribution may also face elevated risks in certain contexts.[196] In principle, design defect doctrine could be extended to address such disparities. This Article does not pursue that broader project. Instead, it advances a limited and doctrinally grounded claim: when manufacturers market products as gender neutral while designing them around male bodies, courts should treat the resulting, foreseeable safety disparities as relevant to design defect analysis.

That focus reflects both doctrinal coherence and institutional restraint. Sex-based disparities occupy a distinctive position in law, not because sex is the only relevant axis of difference, but because formally neutral standards have long embedded male norms as universal baselines. Across multiple legal domains, courts and legislatures have recognized that such baselines can systematically disadvantage women even absent discriminatory intent.[197] Products liability doctrine, which asks whether products are reasonably safe for their intended users, has yet to reckon with this same structural dynamic.

Male-default product design presents a closely analogous problem. Products are routinely engineered using male anthropometric and biomechanical data and then marketed as gender neutral, effectively treating male physiology as the design reference point and women's bodies as deviations. The resulting safety disparities are not speculative or incidental; they are the foreseeable consequence of baseline design choices that privilege one group of ordinary users while shifting risk onto another. Recognizing these disparities as legally salient does not require courts to impose a universal design mandate or to second-

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196 Automobiles are an example of one such product. *See* Kahane, *supra* note 3.

197 *See, e.g.*, Equal Pay Act of 1963, 29 U.S.C. § 206(d) (addressing sex-based disparities embedded in ostensibly neutral compensation practices); Title VII of the Civil Rights Act of 1964, 42 U.S.C. § 2000e-2(a) (prohibiting employment practices that have disparate effects on the basis of sex absent business necessity); Title IX of the Education Amendments of 1972, 20 U.S.C. § 1681(a) (prohibiting sex-based exclusion through formally neutral institutional rules); *see also* Cal. Fed. Sav. & Loan Ass'n v. Guerra, 479 U.S. 272, 289–90 (1987) (recognizing that formally neutral policies may perpetuate historical disadvantage for women); Griggs v. Duke Power Co., 401 U.S. 424, 431 (1971) (recognizing disparate-impact liability where neutral standards embed group-based disadvantage).

guess every design tradeoff. It requires only the modest proposition that products marketed to women as well as men should be reasonably safe for both.

CONCLUSION

Products liability law traditionally asks whether a product is reasonably safe. This Article contends that the inquiry is incomplete unless courts also ask *for whom*. When products are designed around male bodies, marketed as gender neutral, and foreseeably expose women to heightened risks of injury, those harms are not anomalous; they are the predictable consequence of male-default design. Yet prevailing design defect doctrine, with its emphasis on aggregate safety outcomes, has largely failed to register these disparities.

This failure is not compelled by existing law. Both the risk–utility test and the consumer expectations test permit courts to consider whether a product is defective for a substantial and foreseeable class of ordinary users. Courts have occasionally done so in cases involving children and allergies, but have rarely extended the same reasoning to sex-based design baselines. As a result, products that perform well for men but systematically endanger women are often insulated from liability.

Incorporating subgroup considerations into design defect analysis would correct this doctrinal blind spot. At a minimum, courts should ensure that aggregate measures of safety do not obscure predictable harms to women. More fully, persistent sex-based safety disparities resulting from male-centered design choices should be treated as relevant to defectiveness. Recognizing such disparities would better align products liability law with its core commitments to fairness and deterrence—and ensure that products marketed to all users are reasonably safe for those to whom they are sold.